

PC-Part-Picker

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1. Einführung - Funktion der Applikation

- Konfigurieren von PC's
- Kompatibilitätsprüfung der Komponenten
- Speichern von Entwürfen und fertigen PC's
- Benchmarks durchführen

1.1 Start der Applikation

- Voraussetzungen:
 - Java 17 oder höher
 - Maven installiert
- Terminal im „tinf23b3-pcpartpicker“ Ordner öffnen
- Startbefehl (Maven):

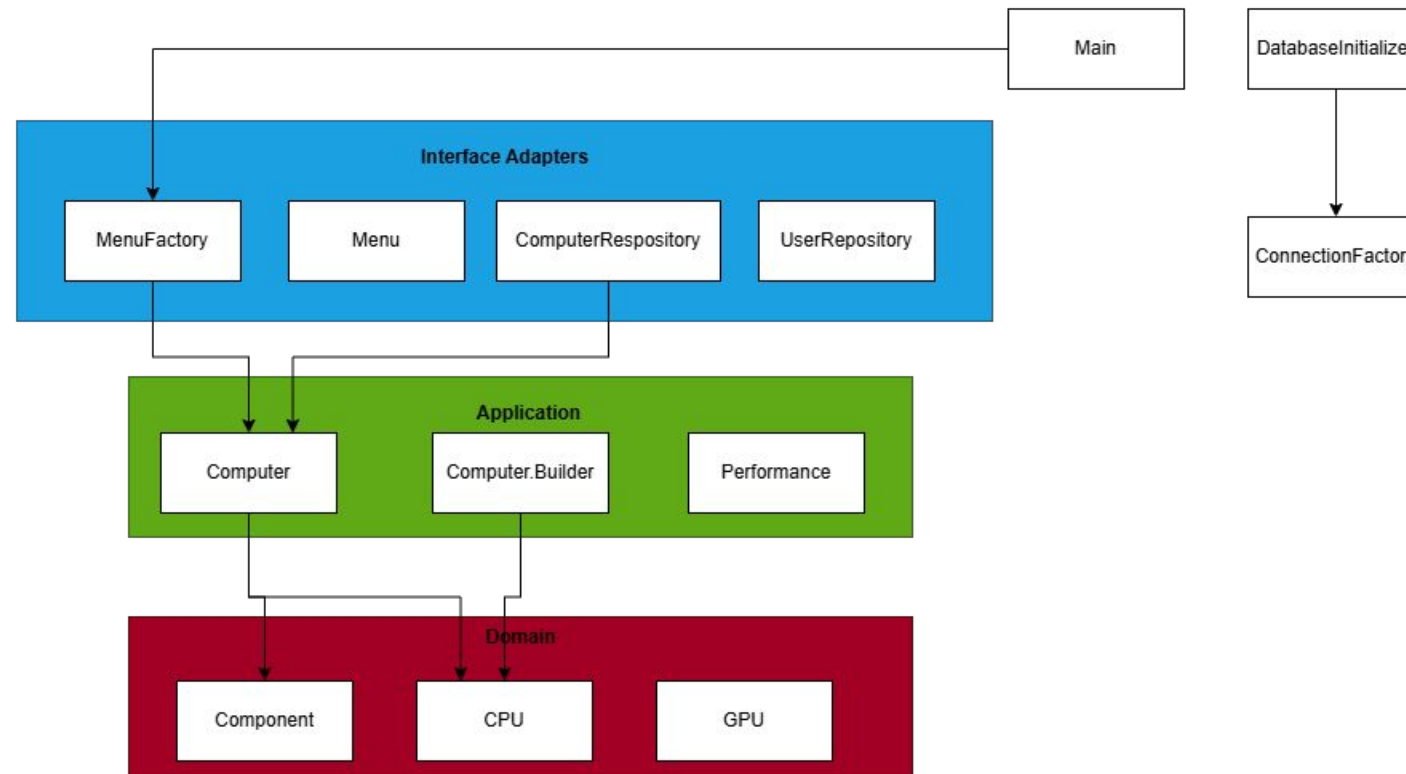


```
1 mvn exec:java "-Dexec.mainClass=de.ase.pcpartpicker.Main"
```

1.2 Technischer Überblick

- Programmiersprache: Java
- Build-Tool: Maven
- Datenbankmanagement: SQLite
- Einlesen der Seed-Daten (JSONL): Gson
- Testen: Junit 5

2. Softwarearchitektur – Clean Architecture



2.1 Domain Code

- Repräsentiert reines Fachwissen ohne Abhängigkeiten

CPU.java

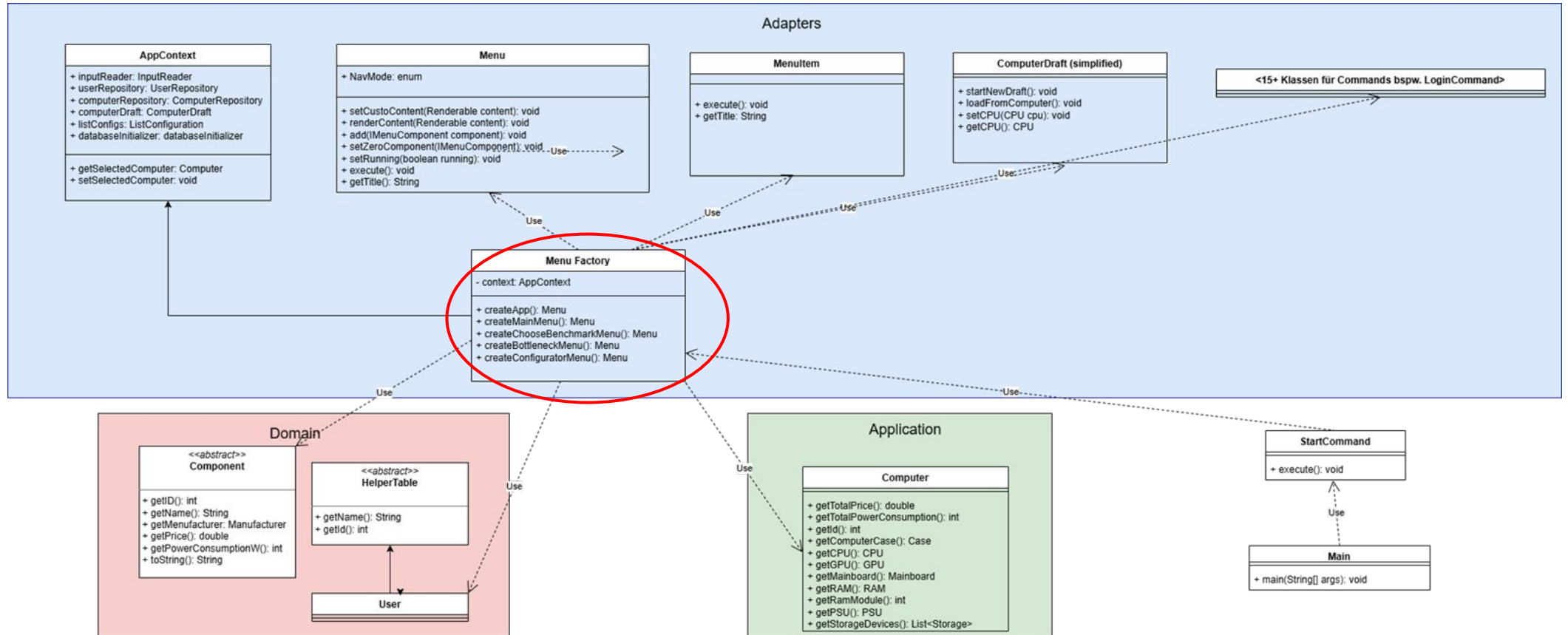
```
1  /**
2   * Klasse für die CPU-Komponente.
3   * @param id Eindeutige ID der CPU
4   * @param name Name der CPU
5   * @param price Preis der CPU
6   * @param manufacturer Hersteller der CPU
7   * @param socket Sockeltyp der CPU (z.B. AM4, LGA1200)
8   * @param speedGHz Taktfrequenz der CPU in GHz
9   * @param boostClockGHz optionale Boost-Frequenz der CPU in GHz
10  * @param hasIntegratedGraphics Gibt an, ob die CPU integrierte Grafik hat
11  * @param powerConsumptionW Maximaler Stromverbrauch der Komponente in Watt
12  * @author Fabio
13  */
14  public class CPU extends Component {
15
16      private final Socket socket;
17      private final double speedGHz;
18      private final Double boostClockGHz;
19      private final boolean hasIntegratedGraphics;
20
21      public CPU(int id, String name, double price, Manufacturer manufacturer, Socket socket, double speedGHz, boolean hasIntegratedGraphics, int powerConsumptionW) {
22          this(id, name, price, manufacturer, socket, speedGHz, null, hasIntegratedGraphics, powerConsumptionW);
23      }
24
25      public CPU(int id, String name, double price, Manufacturer manufacturer, Socket socket, double speedGHz, Double boostClockGHz, boolean hasIntegratedGraphics, int powerConsumptionW) {
26          super(id, name, price, manufacturer, powerConsumptionW);
27          this.socket = socket;
28          this.speedGHz = speedGHz;
29          this.boostClockGHz = boostClockGHz;
30          this.hasIntegratedGraphics = hasIntegratedGraphics;
31      }
32
33      public Socket getSocket() {
34          return socket;
35      }
36
37      public double getSpeedGHz() {
38          return speedGHz;
39      }
40
41      public Double getBoostClockGHz() {
42          return boostClockGHz;
43      }
44
45      public boolean hasIntegratedGraphics() {
46          return hasIntegratedGraphics;
47      }
48  }
```

2.2 Dependency Rule - Positiv

MenuFactory.java

```
1 package de.ase.pcpartpicker.adapters.cli;
2
3 import java.util.List;
4 import java.util.function.BiConsumer;
5 import java.util.function.Supplier;
6
7 import de.ase.pcpartpicker.ColorConstants;
8 import de.ase.pcpartpicker.adapters.cli.commands.AutomaticFixCommand;
9 import de.ase.pcpartpicker.adapters.cli.commands.ComputerToBenchmarkCommand;
10 import de.ase.pcpartpicker.adapters.cli.commands.ComputerToCheckCommand;
11 import de.ase.pcpartpicker.adapters.cli.commands.FinishComputerCommand;
12 import de.ase.pcpartpicker.adapters.cli.commands.LoginCommand;
13 import de.ase.pcpartpicker.adapters.cli.commands.LogoutCommand;
14 import de.ase.pcpartpicker.adapters.cli.commands.NewUserCommand;
15 import de.ase.pcpartpicker.adapters.cli.commands.OpenMenuCommand;
16 import de.ase.pcpartpicker.adapters.cli.commands.ResetDatabaseCommand;
17 import de.ase.pcpartpicker.adapters.cli.commands.RunBenchmarkCommand;
18 import de.ase.pcpartpicker.adapters.cli.commands.RunBottleneckCheckCommand;
19 import de.ase.pcpartpicker.adapters.cli.commands.SaveDraftCommand;
20 import de.ase.pcpartpicker.adapters.cli.commands.SelectComponentCommand;
21 import de.ase.pcpartpicker.adapters.cli.commands.ShowAllUserCommand;
22 import de.ase.pcpartpicker.adapters.cli.commands.ShowComputerCommand;
23 import de.ase.pcpartpicker.adapters.cli.commands.ShowCurrentDraftCommand;
24 import de.ase.pcpartpicker.adapters.cli.commands.ShowListCommand;
25 import de.ase.pcpartpicker.adapters.cli.commands.StartComputerDraftCommand;
26 import de.ase.pcpartpicker.adapters.cli.utils.NavigationUtils;
27 import de.ase.pcpartpicker.domain.Component;
28 import de.ase.pcpartpicker.domain.HDD;
29 import de.ase.pcpartpicker.domain.HelperClasses.User;
30 import de.ase.pcpartpicker.domain.M2SSD;
31 import de.ase.pcpartpicker.domain.SSD;
32 import de.ase.pcpartpicker.part_assembly.Computer;
33
34 /**
35  * Klasse, die die Menüs erstellt und konfiguriert.
36  * Bezieht alle Abhängigkeiten über den AppContext.
37  */
38 public class MenuFactory {
39
40
41     private final AppContext context;
42
43     public MenuFactory(AppContext context) {
44         this.context = context;
45     }
46
47     public static Menu createApp() {
48         AppContext context = new AppContext();
49         return new MenuFactory(context).createMainMenu();
50     }
51
52     public Menu createMainMenu() {
53         Menu mainMenu = new Menu("PC Part Picker - Hauptmenü", context.inputReader);
54         mainMenu.add(new MenuItem("Verfügbare Teile ansehen", new OpenMenuCommand(createComponentMenu())));
55         mainMenu.add(new MenuItem("Computerverwaltung", new OpenMenuCommand(createComputerMenu())));
56         mainMenu.add(new MenuItem("Nutzerverwaltung", new OpenMenuCommand(createUserMenu())));
57         mainMenu.add(new MenuItem("Login", new OpenMenuCommand(createLoginMenu())));
58         mainMenu.add(new MenuItem("Datenbank reset", new ResetDatabaseCommand(context.inputReader, context.databaseInitializer)));
59
60         NavigationUtils.addExitNavigation(mainMenu);
61         return mainMenu;
62     }
63 }
```

2.2 Dependency Rule – Positiv - UML



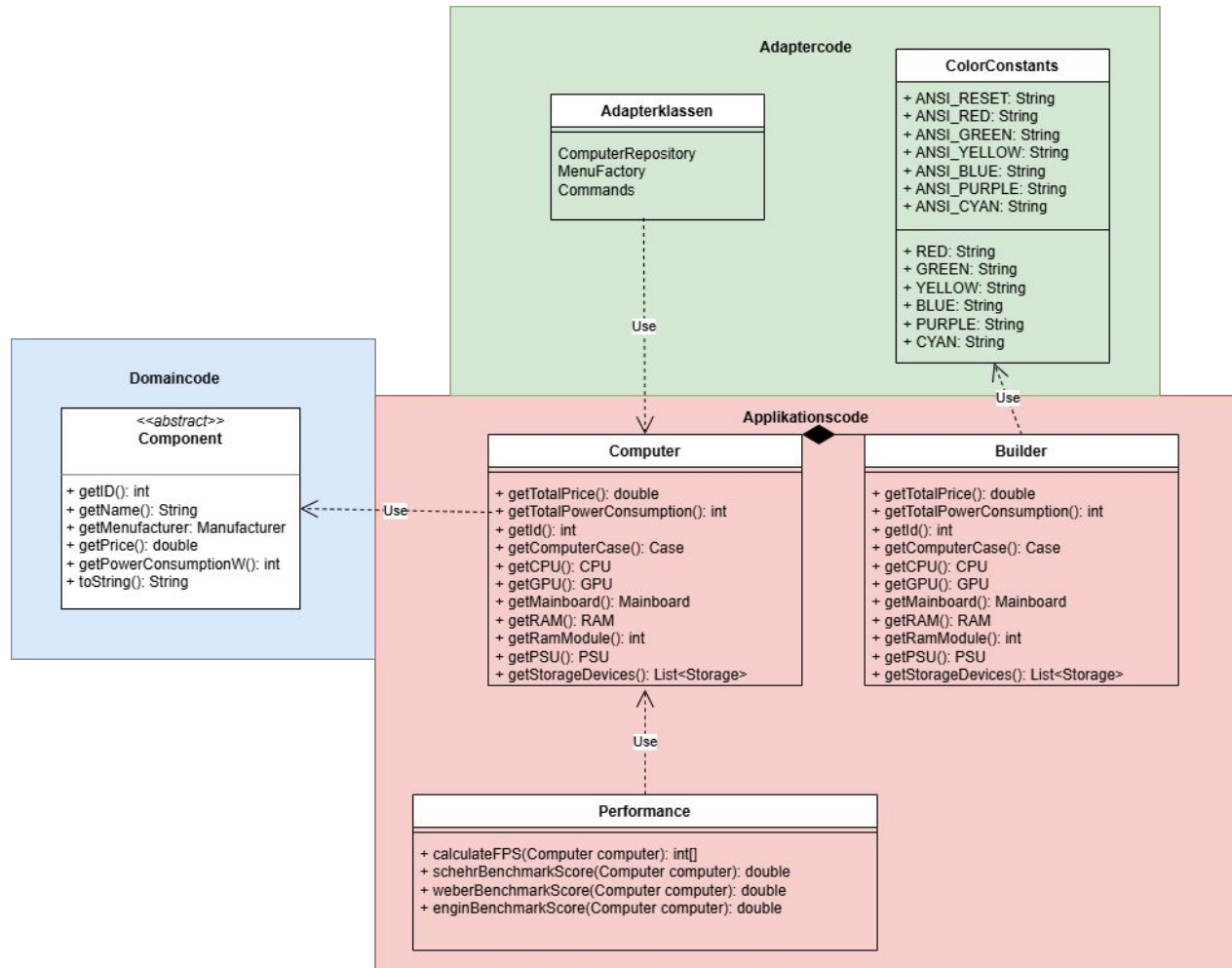
2.2 Dependency Rule - Negativ

Computer.Builder.java

```
1 package de.ase.pcpartpicker.part_assembly;
2
3 import java.util.ArrayList;
4 import java.util.List;
5
6 import de.ase.pcpartpicker.ColorConstants;
7 import de.ase.pcpartpicker.domain.CPU;
8 import de.ase.pcpartpicker.domain.Case;
9 import de.ase.pcpartpicker.domain.Component;
10 import de.ase.pcpartpicker.domain.GPU;
11 import de.ase.pcpartpicker.domain.Mainboard;
12 import de.ase.pcpartpicker.domain.PSU;
13 import de.ase.pcpartpicker.domain.RAM;
14 import de.ase.pcpartpicker.domain.Storage;
15
16 /**
17  * Klasse die einen vollständig konfigurierten Computer abbildet
18  * Diese Klasse ist unveränderlich. <br>
19  * {@link Computer.Builder} dient zum Erstellen von Computer-Objekten
20  *
21  * @author Tuluhan
22  */
23 public class Computer {
24
25     private final int id;
26     private final Case computerCase;
27     private final CPU cpu;
28     private final GPU gpu;
29     private final Mainboard mainboard;
30     private final RAM ram;
31     private final int ramModule; // Anzahl der RAM-Module, z.B. 2 für 2x8GB
32     private final PSU psu;
33     private final List<Storage> storageDevices; // Array von Speichermedien (z.B. SSDs, HDDs)
34
35     /**
36      * Erzeugt einen Computer basierend auf den Einstellungen im Builder.
37      * @param builder Der Builder, der die Komponenten enthält.
38      */
39     private Computer(Builder builder) {
40         this.id = builder.id;
41         this.computerCase = builder.computerCase;
42         this.cpu = builder.cpu;
43         this.gpu = builder.gpu;
44         this.mainboard = builder.mainboard;
45         this.ram = builder.ram;
46         this.ramModule = builder.ramModule;
47         this.psu = builder.psu;
48         this.storageDevices = builder.storageDevices;
49     }
50 }
```

```
1 /**
2  * Diese Methode überprüft die Kompatibilität der Komponenten.
3  * Es wird lediglich überprüft, ob der PC so gebaut werden und existieren könnte.
4  * Bspw. ob das Netzteil genug Leistung hat, um die Komponenten zu versorgen, wird hier nicht überprüft.
5  * @return True, wenn die Komponenten kompatibel sind und der Computer erstellt werden kann, andernfalls false. Es werden auch Fehlermeldungen ausgegeben, die erklären, warum der Computer nicht erstellt werden kann.
6  * @author Tuluhan
7  */
8 private boolean validate() {
9
10     if(ram == null) {
11         System.out.println(ColorConstants.RED("FEHLER") + " | Es muss mindestens ein RAM-Modul ausgewählt werden.");
12         return false;
13     }
14
15     if(cpu == null) {
16         System.out.println(ColorConstants.RED("FEHLER") + " | Es muss eine CPU ausgewählt werden.");
17         return false;
18     }
19
20     if(!cpu.hasIntegratedGraphics() && gpu == null) {
21         System.out.println(ColorConstants.YELLOW("WARNUNG") + " | Der Computer hat keine dedizierte Grafikkarte und die CPU verfügt nicht über integrierte Grafik. Es könnte zu Problemen bei der Anzeige kommen.");
22     }
23
24     if(mainboard == null) {
25         System.out.println(ColorConstants.RED("FEHLER") + " | Es muss ein Mainboard ausgewählt werden.");
26         return false;
27     }
28
29     if(psu == null) {
30         System.out.println(ColorConstants.RED("FEHLER") + " | Es muss ein Netzteil ausgewählt werden.");
31         return false;
32     }
33
34     if(computerCase == null) {
35         System.out.println(ColorConstants.RED("FEHLER") + " | Es muss ein Gehäuse ausgewählt werden.");
36         return false;
37     }
38
39     if(!computerCase.getPSUFormFactor().equals(psu.getFormFactor())) {
40         System.out.println(ColorConstants.RED("FEHLER") + " | Die Form des Netzteils passt nicht zum Gehäuse.");
41         return false;
42     }
43
44     if(!checkMainboardCaseCompatibility(mainboard, computerCase)) {
45         System.out.println(ColorConstants.RED("FEHLER") + " | Das Mainboard passt nicht ins Gehäuse.");
46         return false;
47     }
48
49     if(!cpu.getSocket().equals(mainboard.getSocket())) {
50         System.out.println(ColorConstants.RED("FEHLER") + " | Die CPU ist nicht mit dem Mainboard kompatibel.");
51         return false;
52     }
53
54     if(ramModule > mainboard.getRamSlots()) {
55         System.out.println(ColorConstants.RED("FEHLER") + " | Das Mainboard hat nicht genug RAM-Slots für die Anzahl der RAM-Module.");
56         return false;
57     }
58
59     if(storageDevices.isEmpty()) {
60         System.out.println(ColorConstants.RED("FEHLER") + " | Es muss mindestens ein Speichermedium ausgewählt werden.");
61         return false;
62     }
63
64     if(psu.getWattage() < calculateMomentaryPowerConsumption()) {
65         System.out.println(ColorConstants.RED("FEHLER") + " | Das Netzteil hat nicht genug Leistung um das System zu versorgen. Das Netzteil muss mindestens " + calculateMomentaryPowerConsumption() + "W liefern, um alle Komponenten zu versorgen.");
66         return false;
67     }
68
69     return true;
70 }
```

2.2 Dependency Rule- Negativ- UML

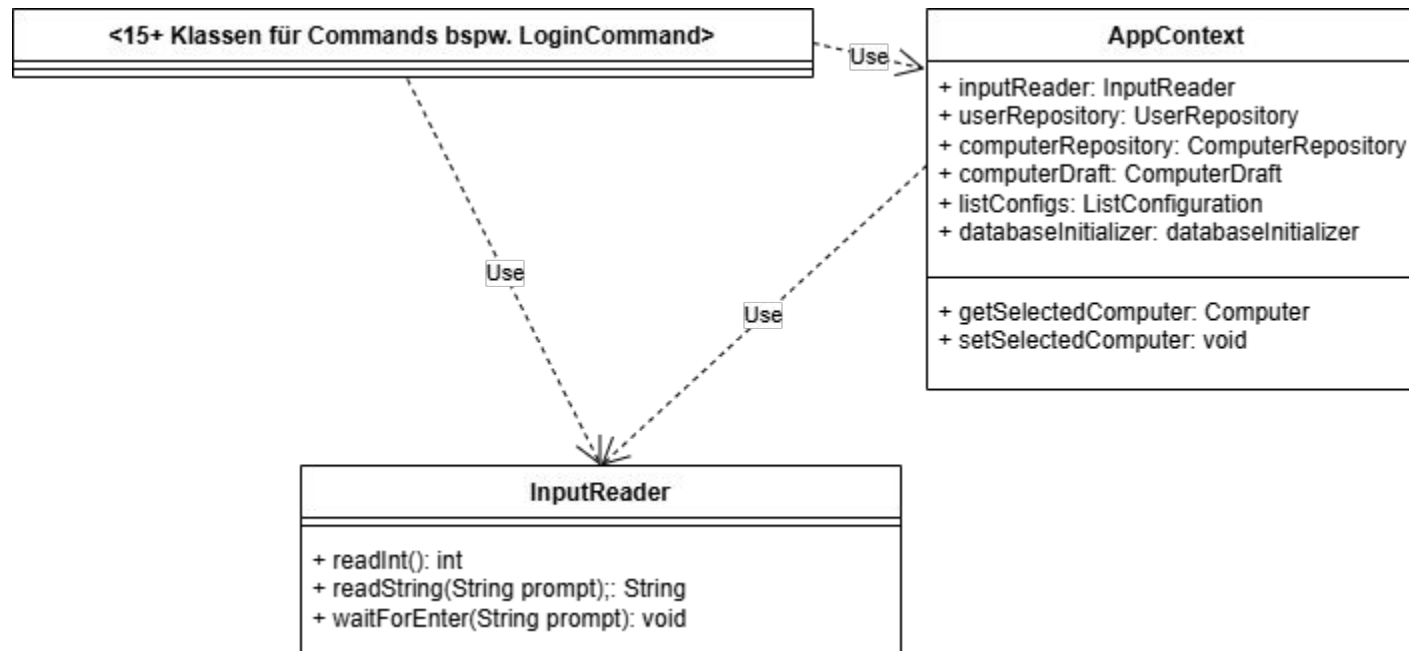


3. SOLID Prinzipien - SRP - Positiv

InputReader.java

```
1 package de.ase.pcpartpicker.adapters.cli;
2
3 import java.util.Scanner;
4
5 /**
6  * Klasse, die Tastatureingaben je nach Datentyp liest
7  * @author Henri
8  */
9
10 public class InputReader {
11     private final Scanner scanner;
12
13     public InputReader() {
14         this.scanner = new Scanner(System.in, "UTF-8");
15     }
16
17
18     public int readInt(String prompt, int min, int max) {
19         int input;
20         while(true){
21             System.out.print(prompt + " (" + min + "-" + max + "): ");
22             try{
23                 input = Integer.parseInt(scanner.nextLine());
24                 if (input >= min && input <= max) return input;
25                 System.out.println("Fehler: Bitte eine Zahl zwischen " + min + " und " + max + " eingeben.");
26             }
27             catch (NumberFormatException e){
28                 System.out.println("Fehler: Ungültige Eingabe. Bitte eine Zahl eingeben.");
29             }
30         }
31     }
32
33
34     public String readString(String prompt) {
35         System.out.println(prompt + ": ");
36         return scanner.nextLine();
37     }
38
39     public void waitForEnter(String prompt) {
40         System.out.println(prompt);
41         scanner.nextLine();
42     }
43 }
```

3. SRP – Positiv – UML



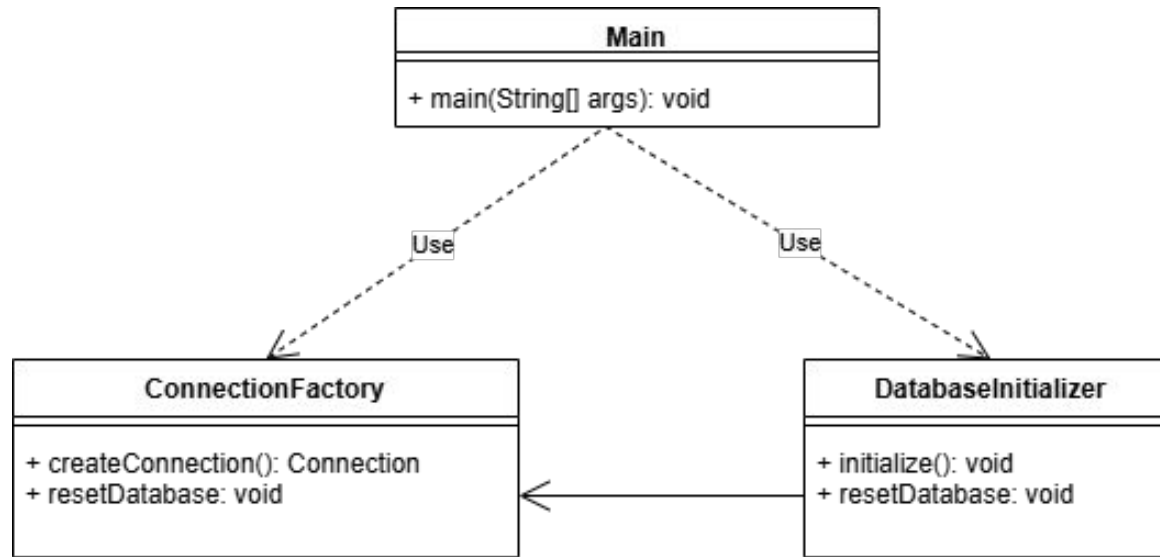
3. SRP – Negativ

DatabaseInitializer.java

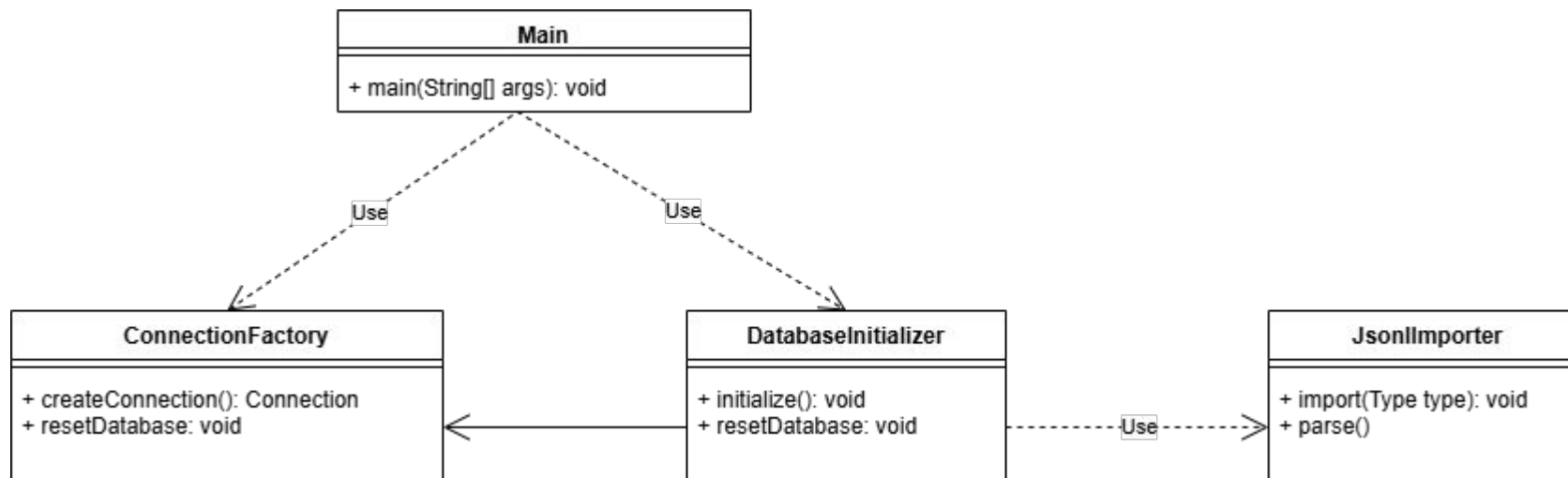
```
1 package de.ase.pcpartpicker.adapters.sqlite;
2
3 import com.google.gson.JsonElement;
4 import com.google.gson.JsonObject;
5 import com.google.gson.JsonParser;
6
7 import java.io.IOException;
8 import java.nio.charset.StandardCharsets;
9 import java.nio.file.Files;
10 import java.nio.file.Path;
11 import java.nio.file.Paths;
12 import java.sql.Connection;
13 import java.sql.PreparedStatement;
14 import java.sql.ResultSet;
15 import java.sql.SQLException;
16 import java.sql.Statement;
17 import java.sql.Types;
18 import java.util.ArrayList;
19 import java.util.List;
20 import java.util.Locale;
21
22 /**
23  * Klasse zur Initialisierung der SQLite-Datenbank.
24  * Erstellt die notwendigen Tabellen, falls sie nicht bereits existieren.
25  * @see DatabaseInitializer#initialize()
26  * @author Fabio
27  */
28 public class DatabaseInitializer {
29
30     private final ConnectionFactory connectionFactory;
31     private final Path dataDirectory;
32
33     public DatabaseInitializer(ConnectionFactory connectionFactory) {
34         this(connectionFactory, null);
35     }
36
37     public DatabaseInitializer(ConnectionFactory connectionFactory, Path dataDirectory) {
38         this.connectionFactory = connectionFactory;
39         this.dataDirectory = dataDirectory;
40     }
41
42     /**
43      * Initialisiert die Datenbank, indem sie die erforderlichen Tabellen erstellt, falls diese nicht bereits existieren.
44      * @throws IllegalStateException, wenn die Initialisierung fehlschlägt.
45      */
46     public void initialize() {
47         try (Connection connection = connectionFactory.createConnection();
48             Statement statement = connection.createStatement()) {
49             statement.execute("PRAGMA foreign_keys = ON");
50
51             createSchema(statement);
52
53             if (areAllComponentTablesEmpty(connection)) {
54                 seedFromJson(connection);
55             }
56         } catch (SQLException e) {
57             throw new IllegalStateException("SQLite-Initialisierung fehlgeschlagen.", e);
58         }
59     }
60 }
```

```
1 private void seedFromJson(Connection connection) {
2     try {
3         int cpuTypeId = ensureNamedId(connection, "type", "CPU");
4         int gpuTypeId = ensureNamedId(connection, "type", "GPU");
5         int ramTypeId = ensureNamedId(connection, "type", "RAM");
6         int mainboardTypeId = ensureNamedId(connection, "type", "Mainboard");
7         int psuTypeId = ensureNamedId(connection, "type", "PSU");
8         int caseTypeId = ensureNamedId(connection, "type", "Case");
9         int hddTypeId = ensureNamedId(connection, "type", "HDD");
10        int ssdTypeId = ensureNamedId(connection, "type", "SSD");
11        int m2SsdTypeId = ensureNamedId(connection, "type", "M2SSD");
12
13        System.out.println("Erstelle CPUs ...");
14        importCpu(connection, cpuTypeId);
15        System.out.println("Erstelle GPUs ...");
16        importGpu(connection, gpuTypeId);
17        System.out.println("Erstelle RAM ...");
18        importRam(connection, ramTypeId);
19        System.out.println("Erstelle Mainboards ...");
20        importMainboard(connection, mainboardTypeId);
21        System.out.println("Erstelle PSUs ...");
22        importPsu(connection, psuTypeId);
23        System.out.println("Erstelle Cases ...");
24        importCase(connection, caseTypeId);
25        System.out.println("Erstelle Speicher ...");
26        importStorage(connection, hddTypeId, ssdTypeId, m2SsdTypeId);
27        System.out.println("Import abgeschlossen.");
28    } catch (SQLException e) {
29        throw new IllegalStateException("Seed aus JSONL fehlgeschlagen.", e);
30    }
31 }
32
33 private void importCpu(Connection connection, int typeId) throws SQLException {
34     String sql = """
35         INSERT INTO cpu (manufacturer_id, type_id, name, price, socket_id, speed_ghz, boost_clock, hasIntegratedGraphics, power_consumption_w)
36         VALUES (?, ?, ?, ?, ?, ?, ?, ?);
37     """;
38
39     importRows(connection, "cpu.jsonl", sql, (row, ps) -> {
40         String name = getString(row, "name");
41         Double price = getDouble(row, "price");
42         Double speed = getDouble(row, "core_clock");
43         Double boost = getDouble(row, "boost_clock");
44         Integer tdp = getInt(row, "tdp");
45         if (anyNull(name, price, speed, tdp)) {
46             return false;
47         }
48
49         String manufacturer = extractManufacturer(name);
50         String cleanName = stripManufacturerPrefix(name);
51         String socket = inferCpuSocket(name, getString(row, "microarchitecture"));
52         boolean hasIgpu = isJsonNullOrMissing(row, "graphics");
53
54         int manufacturerId = ensureNamedId(connection, "manufacturer", manufacturer);
55         int socketId = ensureNamedId(connection, "sockets", socket);
56
57         ps.setInt(1, manufacturerId);
58         ps.setInt(2, typeId);
59         ps.setString(3, cleanName);
60         ps.setDouble(4, price);
61         ps.setInt(5, socketId);
62         ps.setDouble(6, speed);
63         if (boost == null) {
64             ps.setNull(7, Types.REAL);
65         } else {
66             ps.setDouble(7, boost);
67         }
68         ps.setBoolean(8, hasIgpu);
69         ps.setInt(9, tdp);
70         return true;
71     });
72 }
```

3. SRP – Negativ - UML



3. SRP – Negativ – UML - Lösung



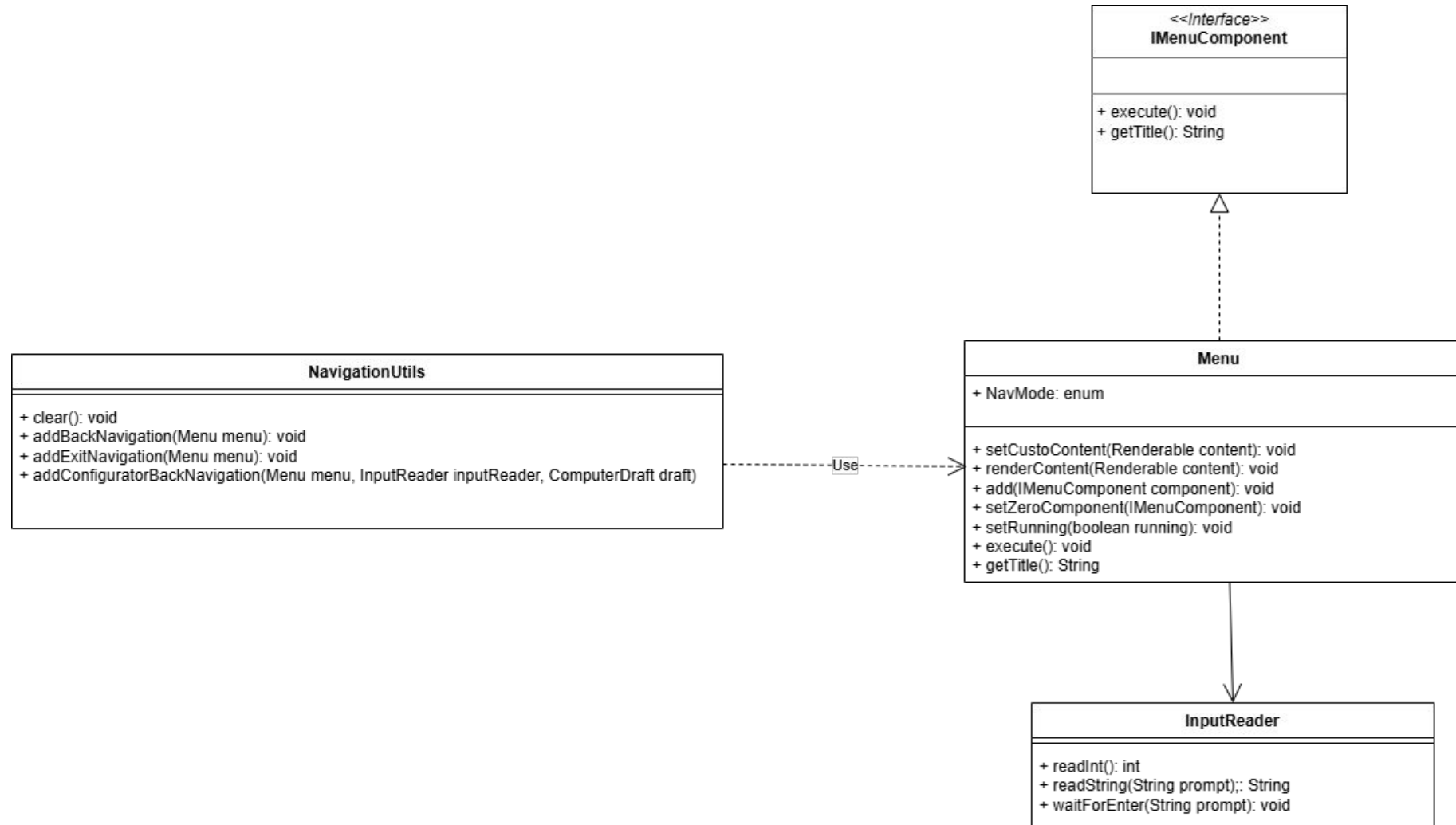
3.1 Open/Close Principle – Positiv

Menu.java

```
1 package de.ase.pcpartpicker.adapters.cli;
2
3 import java.util.ArrayList;
4 import java.util.List;
5
6 import de.ase.pcpartpicker.adapters.cli.utils.NavigationUtils;
7 /**
8  * Klasse, die den grundsätzlichen Aufbau eines Menüs definiert
9  * @param title Titel des Menüs
10  * @param children Enthält Untermenüs, die im aktuellen Menü angezeigt werden können
11  * @param inputReader liest die Benutzereingaben von der Konsole
12  * @param running Flag das angibt, ob das Menü gerade aktiv ist
13  * @author Henri
14  */
15 public class Menu implements IMenuComponent {
16     public enum NavMode {STANDARD, PAGING};
17
18     private final String title;
19     private final List<IMenuComponent> children = new ArrayList<>();
20     private IMenuComponent zeroComponent;
21     private final InputReader inputReader;
22     private boolean running = true;
23     private Renderable customContent;
24     private final NavMode nav;
25
26
27
28     public Menu(String title, InputReader inputReader, NavMode nav) {
29         this.title = title;
30         this.inputReader = inputReader;
31         this.nav = nav;
32     }
33
34     public Menu(String title, InputReader inputReader) {
35         // setze den Standard navigationsmodus
36         this(title, inputReader, NavMode.STANDARD);
37     }
38
39
40     /**
41     * Setzt beliebigen renderbaren Inhalt, der im Menü angezeigt werden soll
42     */
43     public void setCustomContent(Renderable content) {
44         this.customContent = content;
45     }
```

```
1 /**
2  * Gibt beliebigen renderbaren Inhalt direkt im Menü aus
3  */
4     public void renderContent(Renderable content) {
5         if (content != null) {
6             content.render(title);
7         }
8     }
9
10     public void add(IMenuComponent component) {
11         children.add(component);
12     }
13
14     public void setZeroComponent(IMenuComponent component) {
15         this.zeroComponent = component;
16     }
17
18     public void setRunning(boolean running) {
19         this.running = running;
20     }
21
22     @Override
23     public void execute() {
24         running = true;
25         while (running) {
26
27
28             if (nav == NavMode.PAGING && customContent != null) {
29                 customContent.render(title);
30                 running = false;
31                 return;
32             }
33
34             NavigationUtils.clear();
35             System.out.println("\n=== " + title + " ===\n");
36
37             if (SessionManager.isLoggedIn()) {
38                 System.out.println("Eingeloggt als: " + SessionManager.getCurrentUser().getName() + "\n");
39             }
40
41             if (customContent != null) {
42                 customContent.render(title);
43             }
44
45             for (int i = 0; i < children.size(); i++) {
46                 System.out.println((i + 1) + ") " + children.get(i).getTitle());
47             }
48
49             if (zeroComponent != null) {
50                 System.out.println("0) " + zeroComponent.getTitle());
51             }
52
53             System.out.print("\nAuswahl: ");
54
55             int choice = inputReader.readInt("Eine Zahl eingeben", 0, children.size());
56
57
58             if (choice == 0 && zeroComponent != null) {
59                 zeroComponent.execute();
60             } else if (choice > 0 && choice <= children.size()) {
61                 children.get(choice - 1).execute();
62             }
63
64         }
65     }
66
67     @Override
68     public String getTitle() {
69         return title;
70     }
71
72
73 }
```

3.1 Open/Close Principle – Positiv - UML



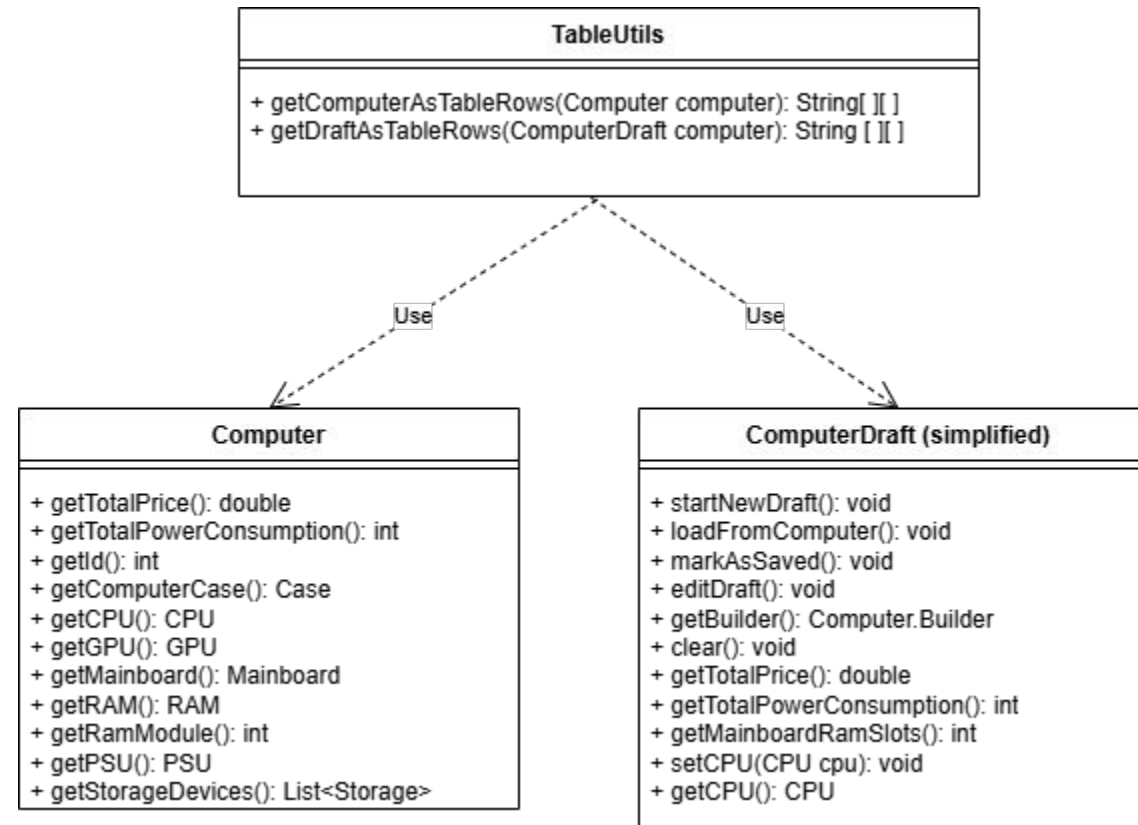
3.1 Open/Close Principle – Negativ

TableUtils.java

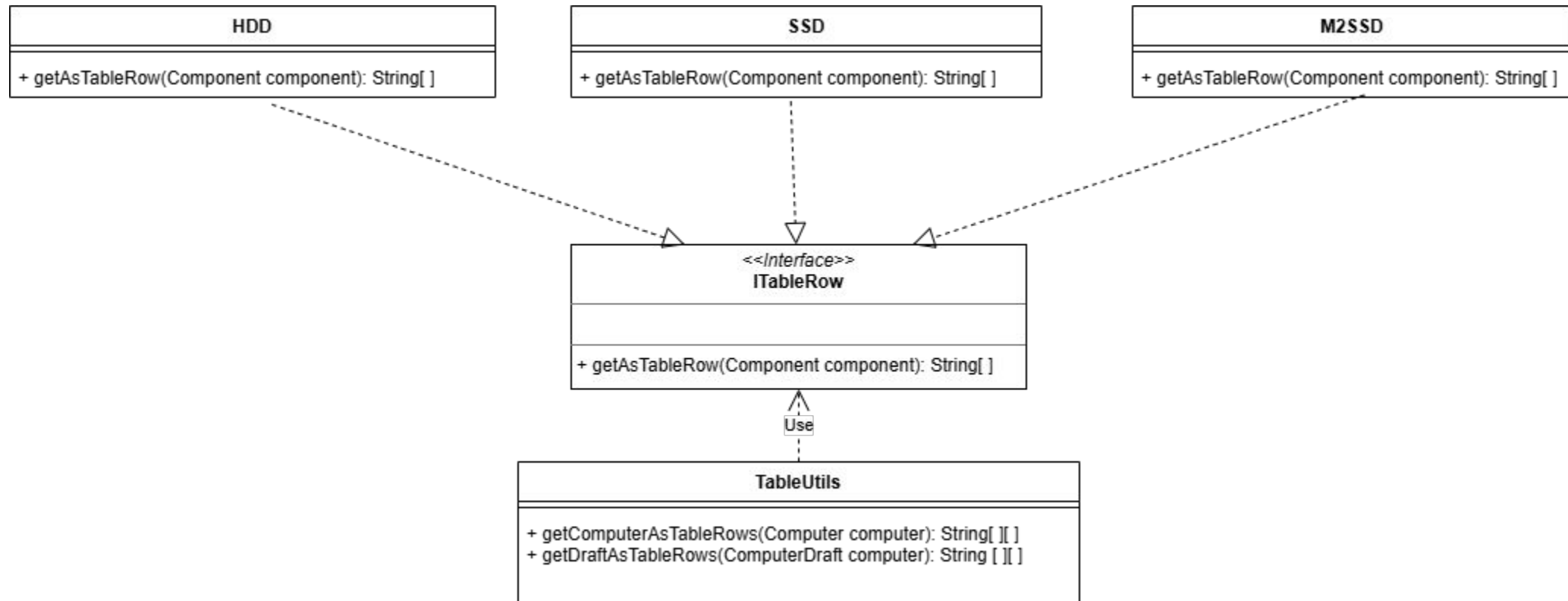
```
1 package de.ase.pcpartpicker.adapters.cli.utils;
2
3 import java.util.List;
4
5 import de.ase.pcpartpicker.adapters.cli.ComputerDraft;
6 import de.ase.pcpartpicker.domain.CPU;
7 import de.ase.pcpartpicker.domain.Case;
8 import de.ase.pcpartpicker.domain.GPU;
9 import de.ase.pcpartpicker.domain.Mainboard;
10 import de.ase.pcpartpicker.domain.PSU;
11 import de.ase.pcpartpicker.domain.RAM;
12 import de.ase.pcpartpicker.domain.Storage;
13 import de.ase.pcpartpicker.part_assembly.Computer;
14
15 public class TableUtils {
16
17     public static String[][] getComputerAsTableRows(Computer computer) {
18         return buildTableRows(
19             computer.getComputerCase(), computer.getCPU(), computer.getGPU(),
20             computer.getMainboard(), computer.getRAM(), computer.getRamModule(),
21             computer.getPSU(), computer.getStorage(),
22             computer.getTotalPowerConsumption(), computer.getTotalPrice()
23         );
24     }
25
26     // 2. Methode für den unfertigen ComputerDraft
27     public static String[][] getDraftAsTableRows(ComputerDraft draft) {
28         return buildTableRows(
29             draft.getComputerCase(), draft.getCPU(), draft.getGPU(),
30             draft.getMainboard(), draft.getRAM(), draft.getRamModule(),
31             draft.getPSU(), draft.getStorage(),
32             draft.getTotalPowerConsumption(), draft.getTotalPrice()
33         );
34     }
35
36     private static String[][] buildTableRows(Case pcCase, CPU cpu, GPU gpu, Mainboard mb, RAM ram, int ramModule, PSU psu, List<Storage> storage, int totalPower, double totalPrice) {
37         List<String> rows = new java.util.ArrayList<>();
38
39         // Gehäuse
40         if (pcCase != null) {
41             rows.add(new String[]{"Gehäuse", "Name", pcCase.getName()});
42             rows.add(new String[]{"", "Formfaktor", pcCase.getMotherboardFormfactor().getName()});
43             rows.add(new String[]{"", "Preis", FormatUtils.formatPrice(pcCase.getPrice())});
44         } else {
45             rows.add(new String[]{"Gehäuse", "nicht ausgewählt", ""});
46         }
47         rows.add(new String[]{"", "", ""});
48
49         // CPU
50         if (cpu != null) {
51             rows.add(new String[]{"CPU", "Name", cpu.getName()});
52             rows.add(new String[]{"", "Kerne", FormatUtils.formatNumber(cpu.getCoreCount())});
53             rows.add(new String[]{"", "Takt", FormatUtils.formatNumber(cpu.getSpeedMHz()) + " MHz"}); // Tippfehler bei dir korrigiert (formatNumber -> formatNumber)
54             rows.add(new String[]{"", "Preis", FormatUtils.formatPrice(cpu.getPrice())});
55         } else {
56             rows.add(new String[]{"CPU", "nicht ausgewählt", ""});
57         }
58         rows.add(new String[]{"", "", ""});
59
60         // GPU
61         if (gpu != null) {
62             rows.add(new String[]{"GPU", "Name", gpu.getName()});
63             rows.add(new String[]{"", "VRAM", FormatUtils.formatNumber(gpu.getVramGB()) + " GB"});
64             rows.add(new String[]{"", "Preis", FormatUtils.formatPrice(gpu.getPrice())});
65         } else {
66             rows.add(new String[]{"GPU", "nicht ausgewählt", ""});
67         }
68         rows.add(new String[]{"", "", ""});
69
70         // Mainboard
71         if (mb != null) {
72             rows.add(new String[]{"Mainboard", "Name", mb.getName()});
73             rows.add(new String[]{"", "Formfaktor", mb.getFormfactor().getName()});
74             rows.add(new String[]{"", "Preis", FormatUtils.formatPrice(mb.getPrice())});
75         } else {
76             rows.add(new String[]{"Mainboard", "nicht ausgewählt", ""});
77         }
78         rows.add(new String[]{"", "", ""});
79
80     }
```

```
1
2 // RAM
3 if (ram != null) {
4     rows.add(new String[]{"RAM", "Name", ram.getName()});
5     rows.add(new String[]{"", "Module", FormatUtils.formatNumber(ramModule)});
6     rows.add(new String[]{"", "Gesamtkapazität", FormatUtils.formatNumber(getRAMCapacity(ramModule, ram.getCapacityGB())) + " GB"});
7     rows.add(new String[]{"", "Takt", FormatUtils.formatNumber(ram.getSpeedMHz()) + " MHz"});
8     rows.add(new String[]{"", "Preis", FormatUtils.formatPrice(ramModule * ram.getPrice())});
9 } else {
10     rows.add(new String[]{"RAM", "nicht ausgewählt", ""});
11 }
12 rows.add(new String[]{"", "", ""});
13
14 // Netzteil
15 if (psu != null) {
16     rows.add(new String[]{"Netzteil", "Name", psu.getName()});
17     rows.add(new String[]{"", "Leistung", FormatUtils.formatNumber(psu.getWattage()) + " W"});
18     rows.add(new String[]{"", "Formfaktor", psu.getFormfactor().getName()});
19     rows.add(new String[]{"", "Preis", FormatUtils.formatPrice(psu.getPrice())});
20 } else {
21     rows.add(new String[]{"Netzteil", "nicht ausgewählt", ""});
22 }
23 rows.add(new String[]{"", "", ""});
24
25 // Storage
26 if (storage != null && !storage.isEmpty()) {
27     int i = 0;
28     for (Storage s : storage) {
29         String comp = (i == 0) ? "Speicher" : "";
30         rows.add(new String[]{"", "Name", s.getName()});
31         rows.add(new String[]{"", "Kapazität", FormatUtils.formatNumber(s.getCapacityGB()) + " GB"});
32         rows.add(new String[]{"", "Preis", FormatUtils.formatPrice(s.getPrice())});
33         rows.add(new String[]{"", "", ""});
34         i++;
35     }
36 } else {
37     rows.add(new String[]{"Speicher", "nicht ausgewählt", ""});
38     rows.add(new String[]{"", "", ""});
39 }
40
41 // Trennlinie und Gesamtpreis (Logik vereint)
42 rows.add(new String[]{"----", "----", "----"});
43 String psuWattage = psu != null ? FormatUtils.formatNumber(psu.getWattage()) + " W" : "0 W";
44 rows.add(new String[]{"Leistungsaufnahme/Gesamt", "", FormatUtils.formatNumber(totalPower) + " W / " + psuWattage});
45 rows.add(new String[]{"Gesamtpreis", "", FormatUtils.formatPrice(totalPrice)});
46
47 return rows.toArray(new String[0][]);
48
49 private static int getRAMCapacity(int module, int capacity) {
50     return module * capacity;
51 }
52
53 }
54
```

3.1 Open/Close Principle – Negativ - UML



3.1 Open/Close Principle – Negativ – UML- Lösung



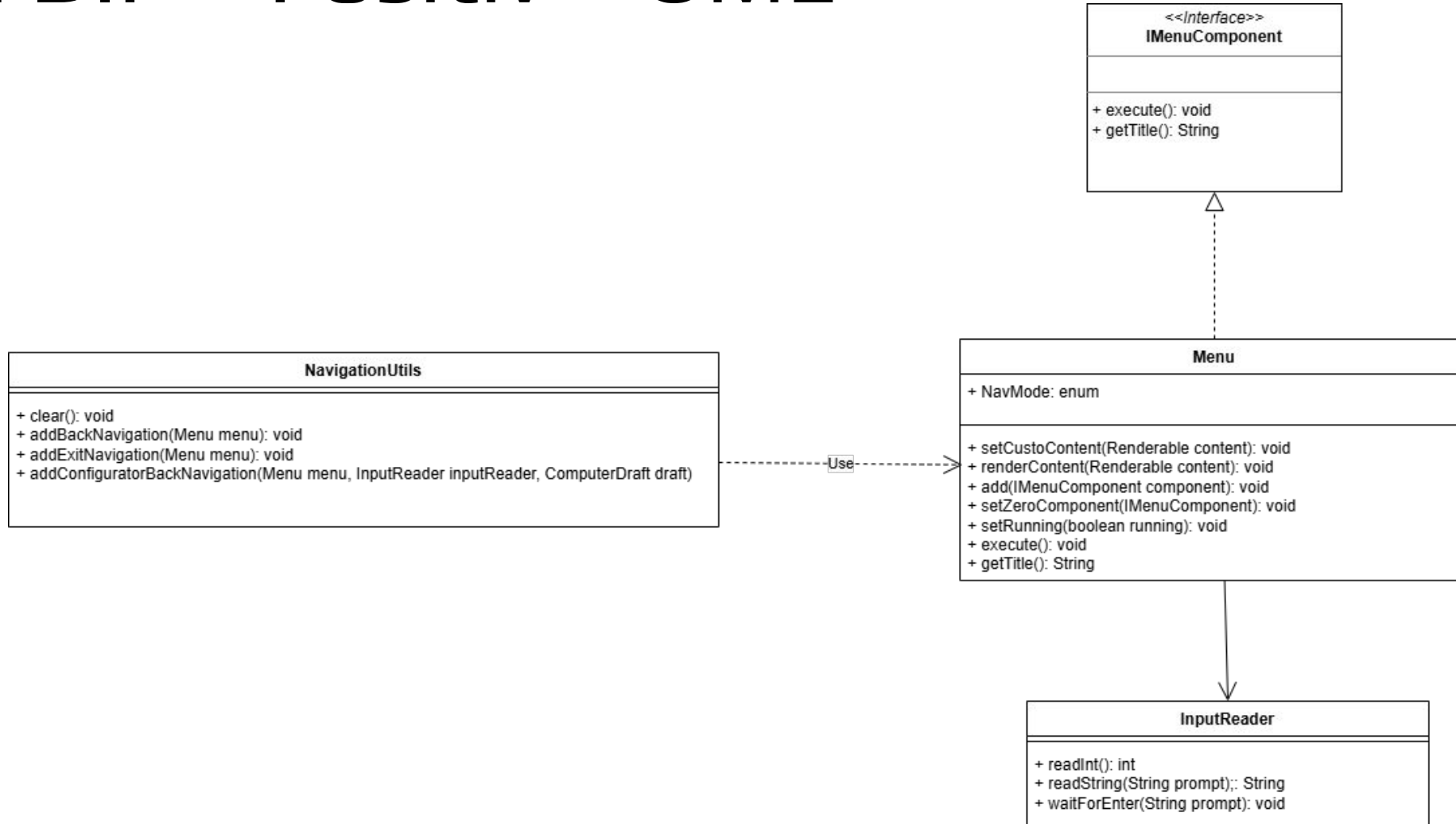
3.2 Dependency Inversion Principle (DIP) - Positiv

Menu.java

```
1 package de.ase.pcpartpicker.adapters.cli;
2
3 import java.util.ArrayList;
4 import java.util.List;
5
6 import de.ase.pcpartpicker.adapters.cli.utils.NavigationUtils;
7 /**
8  * Klasse, die den grundsätzlichen Aufbau eines Menüs definiert
9  * @param title Titel des Menüs
10  * @param children Enthält Untermenüs, die im aktuellen Menü angezeigt werden können
11  * @param inputReader liest die Benutzereingaben von der Konsole
12  * @param running Flag das angibt, ob das Menü gerade aktiv ist
13  * @author Henri
14  */
15 public class Menu implements IMenuComponent {
16     public enum NavMode {STANDARD, PAGING};
17
18     private final String title;
19     private final List<IMenuComponent> children = new ArrayList<>();
20     private IMenuComponent zeroComponent;
21     private final InputReader inputReader;
22     private boolean running = true;
23     private Renderable customContent;
24     private final NavMode nav;
25
26
27     public Menu(String title, InputReader inputReader, NavMode nav) {
28         this.title = title;
29         this.inputReader = inputReader;
30         this.nav = nav;
31     }
32
33     public Menu(String title, InputReader inputReader) {
34         // setze den Standard navigationsmodus
35         this(title, inputReader, NavMode.STANDARD);
36     }
37
38
39     /**
40      * Setzt beliebigen renderbaren Inhalt, der im Menü angezeigt werden soll
41      */
42     public void setCustomContent(Renderable content) {
43         this.customContent = content;
44     }
45 }
```

```
1 /**
2  * Gibt beliebigen renderbaren Inhalt direkt im Menü aus
3  */
4 public void renderContent(Renderable content) {
5     if (content != null) {
6         content.render(title);
7     }
8 }
9
10 public void add(IMenuComponent component) {
11     children.add(component);
12 }
13
14 public void setZeroComponent(IMenuComponent component) {
15     this.zeroComponent = component;
16 }
17
18 public void setRunning(boolean running) {
19     this.running = running;
20 }
21
22 @Override
23 public void execute() {
24     running = true;
25     while (running) {
26
27         if (nav == NavMode.PAGING && customContent != null) {
28             customContent.render(title);
29             running = false;
30             return;
31         }
32
33         NavigationUtils.clear();
34         System.out.println("===== " + title + " =====");
35
36         if (SessionManager.isLoggedIn()) {
37             System.out.println("Eingeloggt als: " + SessionManager.getCurrentUser().getName() + "\n");
38         }
39
40         if (customContent != null) {
41             customContent.render(title);
42         }
43
44         for (int i = 0; i < children.size(); i++) {
45             System.out.println((i + 1) + " " + children.get(i).getTitle());
46         }
47
48         if (zeroComponent != null) {
49             System.out.println("0 " + zeroComponent.getTitle());
50         }
51
52         System.out.print("\nAuswahl: ");
53
54         int choice = inputReader.readInt("Eine Zahl eingeben", 0, children.size());
55
56         if (choice == 0 && zeroComponent != null) {
57             zeroComponent.execute();
58         } else if (choice > 0 && choice <= children.size()) {
59             children.get(choice - 1).execute();
60         }
61     }
62 }
63
64 @Override
65 public String getTitle() {
66     return title;
67 }
68
69 }
```

3.2 DIP – Positiv – UML

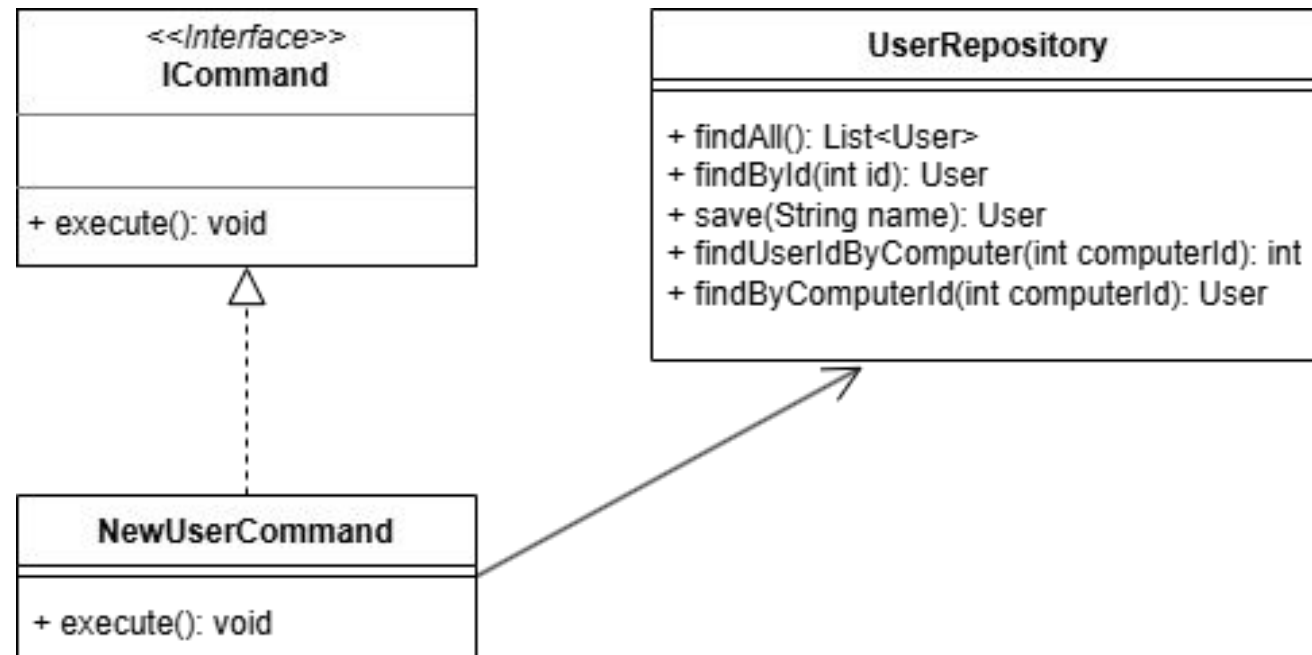


3.2 DIP- Negativ

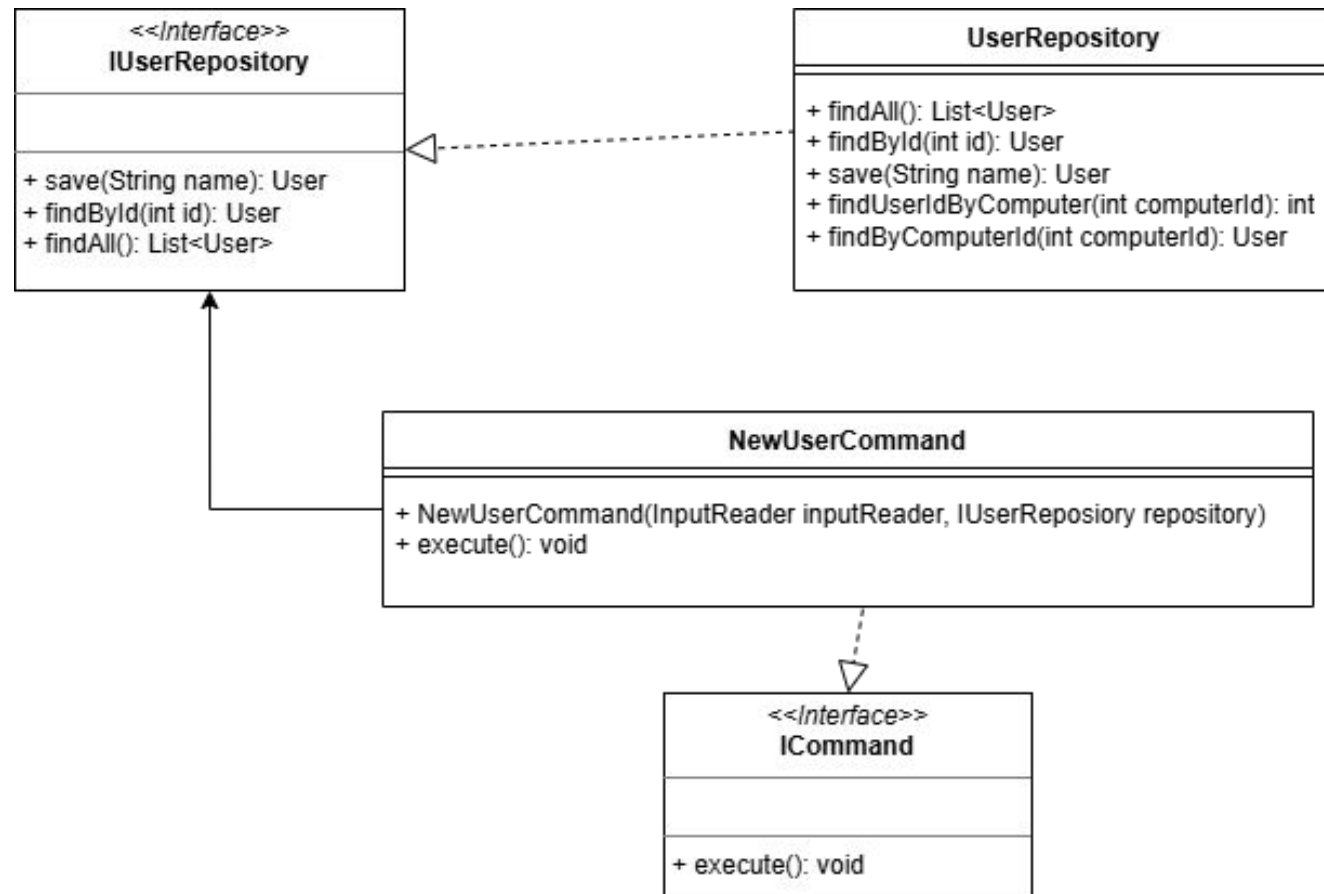
NewUserCommand.java

```
1 package de.ase.pcpartpicker.adapters.cli.commands;
2
3 import de.ase.pcpartpicker.adapters.cli.InputReader;
4 import de.ase.pcpartpicker.adapters.cli.utils.ExceptionUtils;
5 import de.ase.pcpartpicker.adapters.sqlite.repositories.UserRepository;
6
7
8 public class NewUserCommand implements ICommand{
9
10     private final InputReader inputReader;
11     private final UserRepository userRepository;
12
13     public NewUserCommand(InputReader inputReader, UserRepository userRepository) {
14         this.inputReader = inputReader;
15         this.userRepository = userRepository;
16     }
17
18     @Override
19     public void execute() {
20         String username = inputReader.readString("Bitte Nutzernamen eingeben");
21         boolean exists = userRepository.findAll().stream()
22             .anyMatch(user -> user.getName().equalsIgnoreCase(username));
23
24         if (exists) {
25             ExceptionUtils.printInfo("Nutzername existiert bereits!");
26         } else {
27             userRepository.save(username);
28             ExceptionUtils.printInfo("Nutzer erfolgreich angelegt!");
29         }
30
31         inputReader.waitForEnter("\nEnter drücken zum fortfahren...");
32     }
33 }
34
35
36
37 }
38
```

3.2 DIP- Negativ - UML



3.2 DIP- Negativ – UML - Lösung



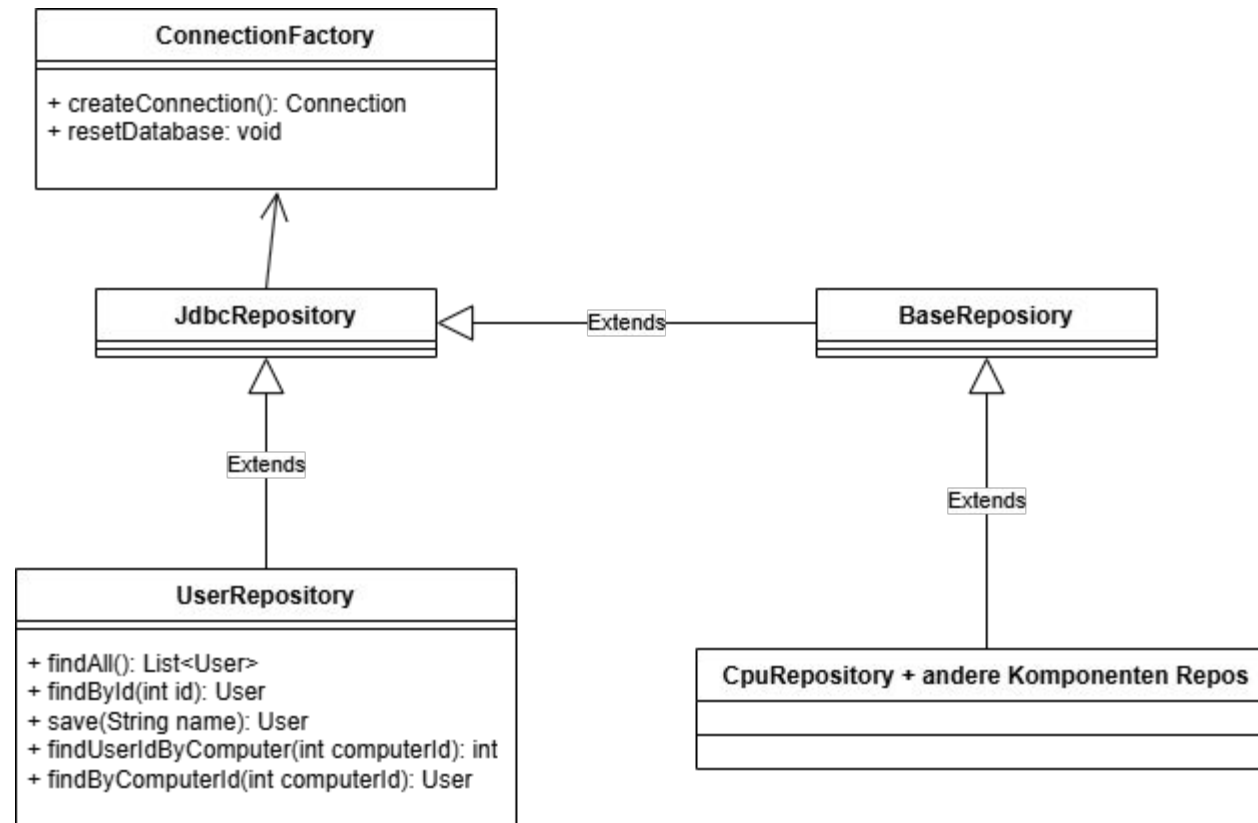
4. GRASP – Low Coupling - Positiv

JdbcRepository.java

```
1 package de.ase.pcpartpicker.adapters.sqlite.repositories;
2
3 import java.sql.Connection;
4 import java.sql.PreparedStatement;
5 import java.sql.ResultSet;
6 import java.sql.SQLException;
7 import java.sql.Statement;
8 import java.util.ArrayList;
9 import java.util.List;
10 import java.util.Optional;
11
12 import de.ase.pcpartpicker.adapters.sqlite.ConnectionFactory;
13
14 /**
15  * Gemeinsame JDBC-Basisklasse für Repositories.
16  *
17  * @param <T> Typ des Domain-Objekts.
18  */
19 public abstract class JdbcRepository<T> implements Repository<T> {
20
21     @FunctionalInterface
22     protected interface RowMapper<R> {
23         R map(ResultSet resultSet) throws SQLException;
24     }
25
26     @FunctionalInterface
27     protected interface StatementConfigurer {
28         void configure(PreparedStatement statement) throws SQLException;
29     }
30
31     protected static final StatementConfigurer NO_PARAMETERS = statement -> {};
32
33     protected final ConnectionFactory connectionFactory;
34
35     protected JdbcRepository(ConnectionFactory connectionFactory) {
36         this.connectionFactory = connectionFactory;
37     }
38
39     protected <R> List<R> queryList(String sql, RowMapper<R> rowMapper, String errorMessage) {
40         return queryList(sql, NO_PARAMETERS, rowMapper, errorMessage);
41     }
42
43     protected <R> List<R> queryList(
44         String sql,
45         StatementConfigurer statementConfigurer,
46         RowMapper<R> rowMapper,
47         String errorMessage
48     ) {
49         List<R> results = new ArrayList<>();
50
51         try (Connection connection = connectionFactory.createConnection();
52             PreparedStatement statement = connection.prepareStatement(sql)) {
53
54             statementConfigurer.configure(statement);
55
56             try (ResultSet resultSet = statement.executeQuery()) {
57                 while (resultSet.next()) {
58                     results.add(rowMapper.map(resultSet));
59                 }
60             }
61             return results;
62
63         } catch (SQLException e) {
64             throw new IllegalStateException(errorMessage, e);
65         }
66     }
67 }
```

```
1 protected <R> Optional<R> queryOptional(String sql, RowMapper<R> rowMapper, String errorMessage) {
2     return queryOptional(sql, NO_PARAMETERS, rowMapper, errorMessage);
3 }
4
5 protected <R> Optional<R> queryOptional(
6     String sql,
7     StatementConfigurer statementConfigurer,
8     RowMapper<R> rowMapper,
9     String errorMessage
10 ) {
11     try (Connection connection = connectionFactory.createConnection();
12         PreparedStatement statement = connection.prepareStatement(sql)) {
13
14         statementConfigurer.configure(statement);
15
16         try (ResultSet resultSet = statement.executeQuery()) {
17             if (resultSet.next()) {
18                 return Optional.empty();
19             }
20             return Optional.of(rowMapper.map(resultSet));
21         }
22     } catch (SQLException e) {
23         throw new IllegalStateException(errorMessage, e);
24     }
25 }
26
27 protected int insertAndReturnGeneratedKey(
28     String sql,
29     StatementConfigurer statementConfigurer,
30     String errorMessage
31 ) {
32     try (Connection connection = connectionFactory.createConnection();
33         PreparedStatement statement = connection.prepareStatement(sql, Statement.RETURN_GENERATED_KEYS)) {
34
35         statementConfigurer.configure(statement);
36         statement.executeUpdate();
37
38         try (ResultSet generatedKeys = statement.getGeneratedKeys()) {
39             if (generatedKeys.next()) {
40                 return generatedKeys.getInt(1);
41             }
42         }
43         throw new IllegalStateException(errorMessage + ": Keine ID zurückgegeben.");
44     } catch (SQLException e) {
45         throw new IllegalStateException(errorMessage, e);
46     }
47 }
48
49 protected int executeUpdate(String sql, StatementConfigurer statementConfigurer, String errorMessage) {
50     try (Connection connection = connectionFactory.createConnection();
51         PreparedStatement statement = connection.prepareStatement(sql)) {
52
53         statementConfigurer.configure(statement);
54         return statement.executeUpdate();
55
56     } catch (SQLException e) {
57         throw new IllegalStateException(errorMessage, e);
58     }
59 }
60
61 }
62 }
```

4. Low Coupling – Positiv – UML

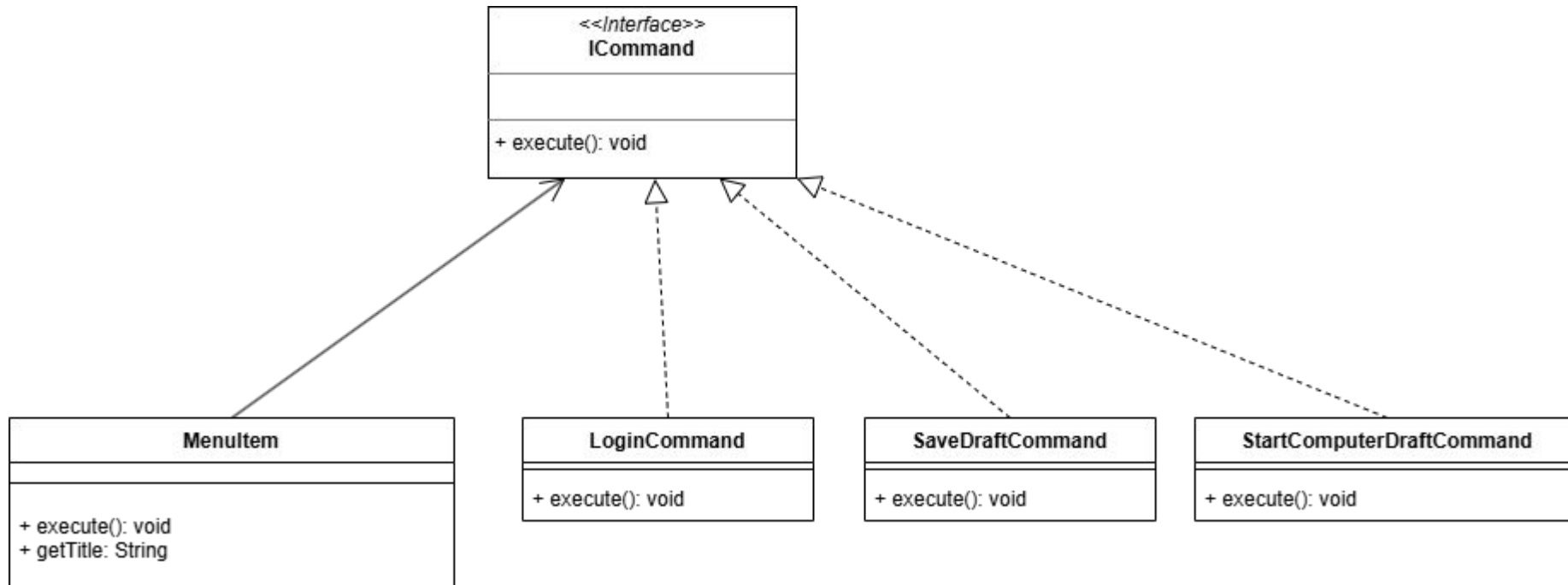


4.1 Polymorphismus - Positiv

MenuItem.java

```
1 package de.ase.pcpartpicker.adapters.cli;
2
3 import java.util.function.Supplier;
4
5 import de.ase.pcpartpicker.adapters.cli.commands.ICommand;
6 /**
7  * Klasse, die definiert welche Komponenten ein Menüeintrag hat
8  * @param title Name des Menüeintrages
9  * @param command Speichert die Aktion die ausgeführt werden soll
10  * @author Henri
11  */
12 public class MenuItem implements IMenuComponent {
13     private final Supplier<String> titleSupplier;
14     private final ICommand command;
15     private Supplier<Boolean> visibilitySupplier = () -> true;
16
17     public MenuItem(String title, ICommand command) {
18         this.titleSupplier = () -> title;
19         this.command = command;
20     }
21
22     // Konstruktor für dynamische Titel
23     public MenuItem(Supplier<String> titleSupplier, ICommand command) {
24         this.titleSupplier = titleSupplier;
25         this.command = command;
26     }
27
28     public MenuItem(String title, ICommand command, Supplier<Boolean> visibilitySupplier) {
29         this.titleSupplier = () -> title;
30         this.command = command;
31         this.visibilitySupplier = visibilitySupplier;
32     }
33
34     @Override
35     public boolean isVisible() {
36         return visibilitySupplier.get();
37     }
38
39     @Override
40     public void execute() {
41         command.execute();
42     }
43
44     @Override
45     public String getTitle() {
46         return titleSupplier.get();
47     }
48 }
```


4.1 Polymorphismus - Positiv - UML



4.2 Dont Repeat Yourself (DRY) - Vorher

TableUtils.java Commit: 72507d2

```
1 package de.unipartitaker.adapters.ii.utils;
2 import java.util.List;
3
4 import de.unipartitaker.adapters.ii.computerinfo;
5 import de.unipartitaker.adapters.ii.storage;
6 import de.unipartitaker.adapters.ii.tableutils;
7
8 public class TableUtils {
9
10     public static String[] getComputerTableData(computer computer) {
11         List<String> rows = new java.util.ArrayList<>();
12
13         // CPU
14         if (computer.getCPU() != null) {
15             rows.add(new String[]{"CPU", "Name", computer.getCPU().getName()});
16             rows.add(new String[]{"", "Faktor", computer.getCPU().getFactor().getCount()});
17             rows.add(new String[]{"", "Preis", computer.getCPU().getPrice()});
18         } else {
19             rows.add(new String[]{"CPU", "nicht ausgwählt", ""});
20         }
21
22         // GPU
23         if (computer.getGPU() != null) {
24             rows.add(new String[]{"GPU", "Name", computer.getGPU().getName()});
25             rows.add(new String[]{"", "Faktor", computer.getGPU().getFactor().getCount()});
26             rows.add(new String[]{"", "Preis", computer.getGPU().getPrice()});
27         } else {
28             rows.add(new String[]{"GPU", "nicht ausgwählt", ""});
29         }
30
31         // RAM
32         if (computer.getRAM() != null) {
33             rows.add(new String[]{"RAM", "Name", computer.getRAM().getName()});
34             rows.add(new String[]{"", "Faktor", computer.getRAM().getFactor().getCount()});
35             rows.add(new String[]{"", "Preis", computer.getRAM().getPrice()});
36         } else {
37             rows.add(new String[]{"RAM", "nicht ausgwählt", ""});
38         }
39
40         // Mainboard
41         if (computer.getMainboard() != null) {
42             rows.add(new String[]{"Mainboard", "Name", computer.getMainboard().getName()});
43             rows.add(new String[]{"", "Faktor", computer.getMainboard().getFactor().getCount()});
44             rows.add(new String[]{"", "Preis", computer.getMainboard().getPrice()});
45         } else {
46             rows.add(new String[]{"Mainboard", "nicht ausgwählt", ""});
47         }
48
49         // Netzteil
50         if (computer.getNetzteil() != null) {
51             rows.add(new String[]{"Netzteil", "Name", computer.getNetzteil().getName()});
52             rows.add(new String[]{"", "Faktor", computer.getNetzteil().getFactor().getCount()});
53             rows.add(new String[]{"", "Preis", computer.getNetzteil().getPrice()});
54         } else {
55             rows.add(new String[]{"Netzteil", "nicht ausgwählt", ""});
56         }
57
58         // Storage
59         if (computer.getStorage() != null) {
60             rows.add(new String[]{"Storage", "Name", computer.getStorage().getName()});
61             rows.add(new String[]{"", "Faktor", computer.getStorage().getFactor().getCount()});
62             rows.add(new String[]{"", "Preis", computer.getStorage().getPrice()});
63         } else {
64             rows.add(new String[]{"Storage", "nicht ausgwählt", ""});
65         }
66
67         // Gesamtsumme
68         if (computer.getTotal() != null) {
69             rows.add(new String[]{"Gesamtsumme", "Name", computer.getTotal().getName()});
70             rows.add(new String[]{"", "Faktor", computer.getTotal().getFactor().getCount()});
71             rows.add(new String[]{"", "Preis", computer.getTotal().getPrice()});
72         } else {
73             rows.add(new String[]{"Gesamtsumme", "nicht ausgwählt", ""});
74         }
75
76         // Gesamtsumme
77         if (computer.getTotal() != null) {
78             rows.add(new String[]{"Gesamtsumme", "Name", computer.getTotal().getName()});
79             rows.add(new String[]{"", "Faktor", computer.getTotal().getFactor().getCount()});
80             rows.add(new String[]{"", "Preis", computer.getTotal().getPrice()});
81         } else {
82             rows.add(new String[]{"Gesamtsumme", "nicht ausgwählt", ""});
83         }
84
85         // Gesamtsumme
86         if (computer.getTotal() != null) {
87             rows.add(new String[]{"Gesamtsumme", "Name", computer.getTotal().getName()});
88             rows.add(new String[]{"", "Faktor", computer.getTotal().getFactor().getCount()});
89             rows.add(new String[]{"", "Preis", computer.getTotal().getPrice()});
90         } else {
91             rows.add(new String[]{"Gesamtsumme", "nicht ausgwählt", ""});
92         }
93
94         // Gesamtsumme
95         if (computer.getTotal() != null) {
96             rows.add(new String[]{"Gesamtsumme", "Name", computer.getTotal().getName()});
97             rows.add(new String[]{"", "Faktor", computer.getTotal().getFactor().getCount()});
98             rows.add(new String[]{"", "Preis", computer.getTotal().getPrice()});
99         } else {
100             rows.add(new String[]{"Gesamtsumme", "nicht ausgwählt", ""});
101         }
102
103         return rows.toArray(new String[0]);
104     }
105
106     public static String[] getTableData(computer computer) {
107         List<String> rows = new java.util.ArrayList<>();
108
109         // CPU
110         if (computer.getCPU() != null) {
111             rows.add(new String[]{"CPU", "Name", computer.getCPU().getName()});
112             rows.add(new String[]{"", "Faktor", computer.getCPU().getFactor().getCount()});
113             rows.add(new String[]{"", "Preis", computer.getCPU().getPrice()});
114         } else {
115             rows.add(new String[]{"CPU", "nicht ausgwählt", ""});
116         }
117
118         // GPU
119         if (computer.getGPU() != null) {
120             rows.add(new String[]{"GPU", "Name", computer.getGPU().getName()});
121             rows.add(new String[]{"", "Faktor", computer.getGPU().getFactor().getCount()});
122             rows.add(new String[]{"", "Preis", computer.getGPU().getPrice()});
123         } else {
124             rows.add(new String[]{"GPU", "nicht ausgwählt", ""});
125         }
126
127         // RAM
128         if (computer.getRAM() != null) {
129             rows.add(new String[]{"RAM", "Name", computer.getRAM().getName()});
130             rows.add(new String[]{"", "Faktor", computer.getRAM().getFactor().getCount()});
131             rows.add(new String[]{"", "Preis", computer.getRAM().getPrice()});
132         } else {
133             rows.add(new String[]{"RAM", "nicht ausgwählt", ""});
134         }
135
136         // Mainboard
137         if (computer.getMainboard() != null) {
138             rows.add(new String[]{"Mainboard", "Name", computer.getMainboard().getName()});
139             rows.add(new String[]{"", "Faktor", computer.getMainboard().getFactor().getCount()});
140             rows.add(new String[]{"", "Preis", computer.getMainboard().getPrice()});
141         } else {
142             rows.add(new String[]{"Mainboard", "nicht ausgwählt", ""});
143         }
144
145         // Netzteil
146         if (computer.getNetzteil() != null) {
147             rows.add(new String[]{"Netzteil", "Name", computer.getNetzteil().getName()});
148             rows.add(new String[]{"", "Faktor", computer.getNetzteil().getFactor().getCount()});
149             rows.add(new String[]{"", "Preis", computer.getNetzteil().getPrice()});
150         } else {
151             rows.add(new String[]{"Netzteil", "nicht ausgwählt", ""});
152         }
153
154         // Storage
155         if (computer.getStorage() != null) {
156             rows.add(new String[]{"Storage", "Name", computer.getStorage().getName()});
157             rows.add(new String[]{"", "Faktor", computer.getStorage().getFactor().getCount()});
158             rows.add(new String[]{"", "Preis", computer.getStorage().getPrice()});
159         } else {
160             rows.add(new String[]{"Storage", "nicht ausgwählt", ""});
161         }
162
163         // Gesamtsumme
164         if (computer.getTotal() != null) {
165             rows.add(new String[]{"Gesamtsumme", "Name", computer.getTotal().getName()});
166             rows.add(new String[]{"", "Faktor", computer.getTotal().getFactor().getCount()});
167             rows.add(new String[]{"", "Preis", computer.getTotal().getPrice()});
168         } else {
169             rows.add(new String[]{"Gesamtsumme", "nicht ausgwählt", ""});
170         }
171
172         // Gesamtsumme
173         if (computer.getTotal() != null) {
174             rows.add(new String[]{"Gesamtsumme", "Name", computer.getTotal().getName()});
175             rows.add(new String[]{"", "Faktor", computer.getTotal().getFactor().getCount()});
176             rows.add(new String[]{"", "Preis", computer.getTotal().getPrice()});
177         } else {
178             rows.add(new String[]{"Gesamtsumme", "nicht ausgwählt", ""});
179         }
180
181         // Gesamtsumme
182         if (computer.getTotal() != null) {
183             rows.add(new String[]{"Gesamtsumme", "Name", computer.getTotal().getName()});
184             rows.add(new String[]{"", "Faktor", computer.getTotal().getFactor().getCount()});
185             rows.add(new String[]{"", "Preis", computer.getTotal().getPrice()});
186         } else {
187             rows.add(new String[]{"Gesamtsumme", "nicht ausgwählt", ""});
188         }
189
190         // Gesamtsumme
191         if (computer.getTotal() != null) {
192             rows.add(new String[]{"Gesamtsumme", "Name", computer.getTotal().getName()});
193             rows.add(new String[]{"", "Faktor", computer.getTotal().getFactor().getCount()});
194             rows.add(new String[]{"", "Preis", computer.getTotal().getPrice()});
195         } else {
196             rows.add(new String[]{"Gesamtsumme", "nicht ausgwählt", ""});
197         }
198
199         // Gesamtsumme
200         if (computer.getTotal() != null) {
201             rows.add(new String[]{"Gesamtsumme", "Name", computer.getTotal().getName()});
202             rows.add(new String[]{"", "Faktor", computer.getTotal().getFactor().getCount()});
203             rows.add(new String[]{"", "Preis", computer.getTotal().getPrice()});
204         } else {
205             rows.add(new String[]{"Gesamtsumme", "nicht ausgwählt", ""});
206         }
207
208         // Gesamtsumme
209         if (computer.getTotal() != null) {
210             rows.add(new String[]{"Gesamtsumme", "Name", computer.getTotal().getName()});
211             rows.add(new String[]{"", "Faktor", computer.getTotal().getFactor().getCount()});
212             rows.add(new String[]{"", "Preis", computer.getTotal().getPrice()});
213         } else {
214             rows.add(new String[]{"Gesamtsumme", "nicht ausgwählt", ""});
215         }
216
217         // Gesamtsumme
218         if (computer.getTotal() != null) {
219             rows.add(new String[]{"Gesamtsumme", "Name", computer.getTotal().getName()});
220             rows.add(new String[]{"", "Faktor", computer.getTotal().getFactor().getCount()});
221             rows.add(new String[]{"", "Preis", computer.getTotal().getPrice()});
222         } else {
223             rows.add(new String[]{"Gesamtsumme", "nicht ausgwählt", ""});
224         }
225
226         // Gesamtsumme
227         if (computer.getTotal() != null) {
228             rows.add(new String[]{"Gesamtsumme", "Name", computer.getTotal().getName()});
229             rows.add(new String[]{"", "Faktor", computer.getTotal().getFactor().getCount()});
230             rows.add(new String[]{"", "Preis", computer.getTotal().getPrice()});
231         } else {
232             rows.add(new String[]{"Gesamtsumme", "nicht ausgwählt", ""});
233         }
234
235         // Gesamtsumme
236         if (computer.getTotal() != null) {
237             rows.add(new String[]{"Gesamtsumme", "Name", computer.getTotal().getName()});
238             rows.add(new String[]{"", "Faktor", computer.getTotal().getFactor().getCount()});
239             rows.add(new String[]{"", "Preis", computer.getTotal().getPrice()});
240         } else {
241             rows.add(new String[]{"Gesamtsumme", "nicht ausgwählt", ""});
242         }
243
244         // Gesamtsumme
245         if (computer.getTotal() != null) {
246             rows.add(new String[]{"Gesamtsumme", "Name", computer.getTotal().getName()});
247             rows.add(new String[]{"", "Faktor", computer.getTotal().getFactor().getCount()});
248             rows.add(new String[]{"", "Preis", computer.getTotal().getPrice()});
249         } else {
250             rows.add(new String[]{"Gesamtsumme", "nicht ausgwählt", ""});
251         }
252
253         // Gesamtsumme
254         if (computer.getTotal() != null) {
255             rows.add(new String[]{"Gesamtsumme", "Name", computer.getTotal().getName()});
256             rows.add(new String[]{"", "Faktor", computer.getTotal().getFactor().getCount()});
257             rows.add(new String[]{"", "Preis", computer.getTotal().getPrice()});
258         } else {
259             rows.add(new String[]{"Gesamtsumme", "nicht ausgwählt", ""});
260         }
261
262         // Gesamtsumme
263         if (computer.getTotal() != null) {
264             rows.add(new String[]{"Gesamtsumme", "Name", computer.getTotal().getName()});
265             rows.add(new String[]{"", "Faktor", computer.getTotal().getFactor().getCount()});
266             rows.add(new String[]{"", "Preis", computer.getTotal().getPrice()});
267         } else {
268             rows.add(new String[]{"Gesamtsumme", "nicht ausgwählt", ""});
269         }
270
271         // Gesamtsumme
272         if (computer.getTotal() != null) {
273             rows.add(new String[]{"Gesamtsumme", "Name", computer.getTotal().getName()});
274             rows.add(new String[]{"", "Faktor", computer.getTotal().getFactor().getCount()});
275             rows.add(new String[]{"", "Preis", computer.getTotal().getPrice()});
276         } else {
277             rows.add(new String[]{"Gesamtsumme", "nicht ausgwählt", ""});
278         }
279
280         // Gesamtsumme
281         if (computer.getTotal() != null) {
282             rows.add(new String[]{"Gesamtsumme", "Name", computer.getTotal().getName()});
283             rows.add(new String[]{"", "Faktor", computer.getTotal().getFactor().getCount()});
284             rows.add(new String[]{"", "Preis", computer.getTotal().getPrice()});
285         } else {
286             rows.add(new String[]{"Gesamtsumme", "nicht ausgwählt", ""});
287         }
288
289         // Gesamtsumme
290         if (computer.getTotal() != null) {
291             rows.add(new String[]{"Gesamtsumme", "Name", computer.getTotal().getName()});
292             rows.add(new String[]{"", "Faktor", computer.getTotal().getFactor().getCount()});
293             rows.add(new String[]{"", "Preis", computer.getTotal().getPrice()});
294         } else {
295             rows.add(new String[]{"Gesamtsumme", "nicht ausgwählt", ""});
296         }
297
298         // Gesamtsumme
299         if (computer.getTotal() != null) {
300             rows.add(new String[]{"Gesamtsumme", "Name", computer.getTotal().getName()});
301             rows.add(new String[]{"", "Faktor", computer.getTotal().getFactor().getCount()});
302             rows.add(new String[]{"", "Preis", computer.getTotal().getPrice()});
303         } else {
304             rows.add(new String[]{"Gesamtsumme", "nicht ausgwählt", ""});
305         }
306
307         // Gesamtsumme
308         if (computer.getTotal() != null) {
309             rows.add(new String[]{"Gesamtsumme", "Name", computer.getTotal().getName()});
310             rows.add(new String[]{"", "Faktor", computer.getTotal().getFactor().getCount()});
311             rows.add(new String[]{"", "Preis", computer.getTotal().getPrice()});
312         } else {
313             rows.add(new String[]{"Gesamtsumme", "nicht ausgwählt", ""});
314         }
315
316         // Gesamtsumme
317         if (computer.getTotal() != null) {
318             rows.add(new String[]{"Gesamtsumme", "Name", computer.getTotal().getName()});
319             rows.add(new String[]{"", "Faktor", computer
```

4.2 DRY - Nachher

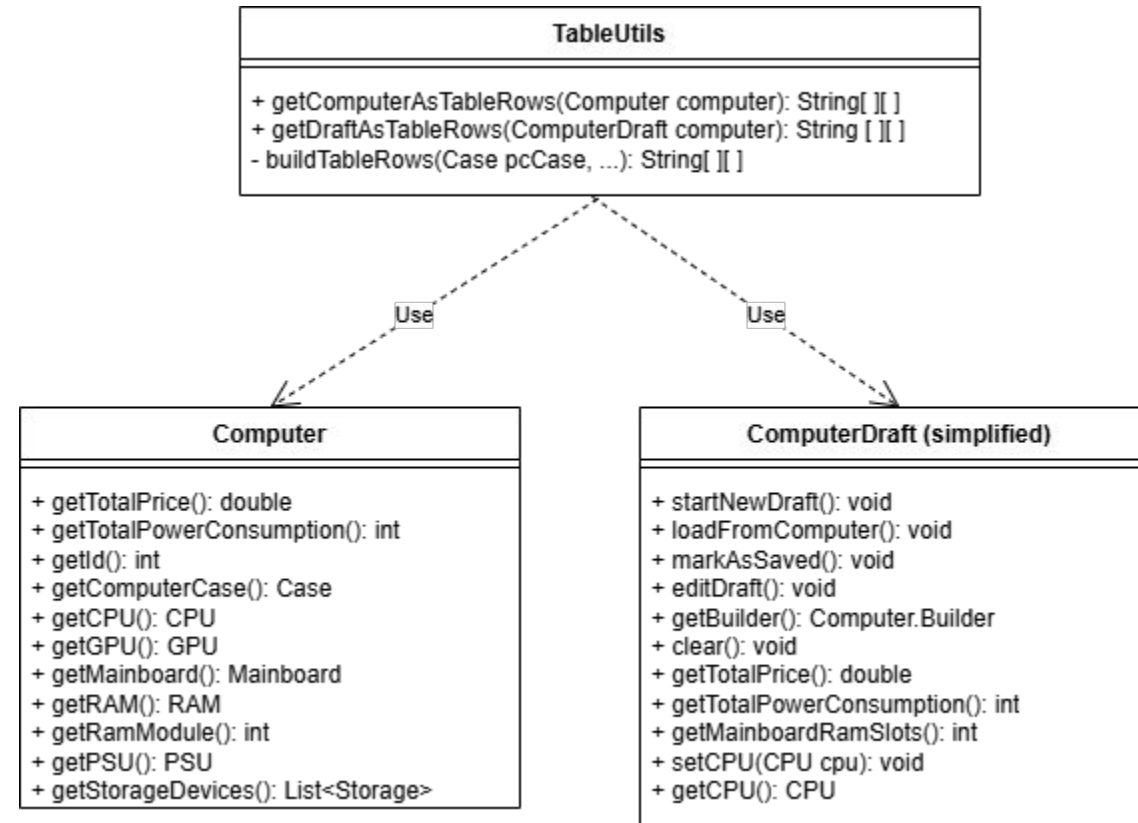
TableUtils.java

Commit: 72507d2

```
1 package de.ase.pcpartpicker.adapters.cli.utils;
2
3 import java.util.List;
4
5 import de.ase.pcpartpicker.adapters.cli.ComputerDraft;
6 import de.ase.pcpartpicker.domain.CPU;
7 import de.ase.pcpartpicker.domain.Case;
8 import de.ase.pcpartpicker.domain.GPU;
9 import de.ase.pcpartpicker.domain.Mainboard;
10 import de.ase.pcpartpicker.domain.PSU;
11 import de.ase.pcpartpicker.domain.RAM;
12 import de.ase.pcpartpicker.domain.Storage;
13 import de.ase.pcpartpicker.part_assembly.Computer;
14
15 public class TableUtils {
16
17     public static String[][] getComputerAsTableRows(Computer computer) {
18         return buildTableRows(
19             computer.getComputerCase(), computer.getCPU(), computer.getGPU(),
20             computer.getMainboard(), computer.getRAM(), computer.getRAMModule(),
21             computer.getPSU(), computer.getStorageDevices(),
22             computer.getTotalPowerConsumption(), computer.getTotalPrice()
23         );
24     }
25
26     // 2. Methode für den unfertigen ComputerDraft
27     public static String[][] getDraftAsTableRows(ComputerDraft draft) {
28         return buildTableRows(
29             draft.getComputerCase(), draft.getCPU(), draft.getGPU(),
30             draft.getMainboard(), draft.getRAM(), draft.getRAMModule(),
31             draft.getPSU(), draft.getStorage(),
32             draft.getTotalPowerConsumption(), draft.getTotalPrice()
33         );
34     }
35
36     private static String[][] buildTableRows(Case pccase, CPU cpu, GPU gpu, Mainboard mb, RAM ram, int ramModule, PSU psu, List<Storage> storage, int totalPower, double totalPrice) {
37         List<String[]> rows = new java.util.ArrayList<>();
38
39         // Gehäuse
40         if (pccase != null) {
41             rows.add(new String[]{"Gehäuse", "Name", pccase.getName()});
42             rows.add(new String[]{"", "Formfaktor", pccase.getMotherboardFormfactor().getName()});
43             rows.add(new String[]{"", "Preis", FormatUtils.formatPrice(pccase.getPrice())});
44         } else {
45             rows.add(new String[]{"Gehäuse", "nicht ausgewählt", ""});
46         }
47         rows.add(new String[]{"", "", ""});
48
49         // CPU
50         if (cpu != null) {
51             rows.add(new String[]{"CPU", "Name", cpu.getName()});
52             rows.add(new String[]{"", "Kerne", FormatUtils.formatNumber(cpu.getCoresCount())});
53             rows.add(new String[]{"", "Takt", FormatUtils.formatNumber(cpu.getSpeedGhz()) + " GHz"}); // Tippfehler bei dir korrigiert (formatNumber -> formatNumber)
54             rows.add(new String[]{"", "Preis", FormatUtils.formatPrice(cpu.getPrice())});
55         } else {
56             rows.add(new String[]{"CPU", "nicht ausgewählt", ""});
57         }
58         rows.add(new String[]{"", "", ""});
59
60         // GPU
61         if (gpu != null) {
62             rows.add(new String[]{"GPU", "Name", gpu.getName()});
63             rows.add(new String[]{"", "VRAM", FormatUtils.formatNumber(gpu.getVramGB()) + " GB"});
64             rows.add(new String[]{"", "Preis", FormatUtils.formatPrice(gpu.getPrice())});
65         } else {
66             rows.add(new String[]{"GPU", "nicht ausgewählt", ""});
67         }
68         rows.add(new String[]{"", "", ""});
69
70         // Mainboard
71         if (mb != null) {
72             rows.add(new String[]{"Mainboard", "Name", mb.getName()});
73             rows.add(new String[]{"", "Formfaktor", mb.getFormfactor().getName()});
74             rows.add(new String[]{"", "Preis", FormatUtils.formatPrice(mb.getPrice())});
75         } else {
76             rows.add(new String[]{"Mainboard", "nicht ausgewählt", ""});
77         }
78         rows.add(new String[]{"", "", ""});
79
80     }
```

```
1
2 // RAM
3 if (ram != null) {
4     rows.add(new String[]{"RAM", "Name", ram.getName()});
5     rows.add(new String[]{"", "Module", FormatUtils.formatNumber(ramModule, ram.getCapacityGB()) + " GB"});
6     rows.add(new String[]{"", "Gesamtkapazität", FormatUtils.formatNumber(ram.getCapacityGB()) + " GB"});
7     rows.add(new String[]{"", "Takt", FormatUtils.formatNumber(ram.getSpeedMhz()) + " MHz"});
8     rows.add(new String[]{"", "Preis", FormatUtils.formatPrice(ramModule * ram.getPrice())});
9 } else {
10     rows.add(new String[]{"RAM", "nicht ausgewählt", ""});
11 }
12 rows.add(new String[]{"", "", ""});
13
14 // Netzteil
15 if (psu != null) {
16     rows.add(new String[]{"Netzteil", "Name", psu.getName()});
17     rows.add(new String[]{"", "Leistung", FormatUtils.formatNumber(psu.getWattage()) + " W"});
18     rows.add(new String[]{"", "Formfaktor", psu.getFormfactor().getName()});
19     rows.add(new String[]{"", "Preis", FormatUtils.formatPrice(psu.getPrice())});
20 } else {
21     rows.add(new String[]{"Netzteil", "nicht ausgewählt", ""});
22 }
23 rows.add(new String[]{"", "", ""});
24
25 // Storage
26 if (storage != null && !storage.isEmpty()) {
27     int i = 0;
28     for (Storage s : storage) {
29         String comp = (i == 0) ? "Speicher" : "";
30         rows.add(new String[]{"", "Name", s.getName()});
31         rows.add(new String[]{"", "Kapazität", FormatUtils.formatNumber(s.getCapacityGB()) + " GB"});
32         rows.add(new String[]{"", "Preis", FormatUtils.formatPrice(s.getPrice())});
33         rows.add(new String[]{"", "", ""});
34         i++;
35     }
36 } else {
37     rows.add(new String[]{"Speicher", "nicht ausgewählt", ""});
38     rows.add(new String[]{"", "", ""});
39 }
40
41 // Trennlinie und Gesamtpreis (Logik vereint)
42 rows.add(new String[]{"---", "---", "---"});
43 String psuWattage = psu != null ? FormatUtils.formatNumber(psu.getWattage()) + " W" : "0 W";
44 rows.add(new String[]{"Leistungsaufnahme/Gesamt", "", FormatUtils.formatNumber(totalPower) + " W / " + psuWattage});
45 rows.add(new String[]{"Gesamtpreis", "", FormatUtils.formatPrice(totalPrice)});
46
47 return rows.toArray(new String[0][]);
48
49 private static int getRAMCapacity(int module, int capacity) {
50     return module * capacity;
51 }
52
53 }
```

4.2 DRY – UML



5. Unit Tests – Menu Tests

```
1  @Test
2  @DisplayName("Testet, ob ShowListCommand eine Liste ausgibt")
3  void showListCommandPrintsList() {
4      ByteArrayOutputStream outContent = new ByteArrayOutputStream();
5      PrintStream originalOut = System.out;
6      System.setOut(new PrintStream(outContent));
7
8      String simulatedInput = "0\n";
9      System.setIn(new java.io.ByteArrayInputStream(simulatedInput.getBytes()));
10
11     ComponentRepository<Component> dummyRepo = new ComponentRepository<Component>() {
12         @Override
13         public java.util.List<Component> findAll() {
14             return java.util.Collections.emptyList();
15         }
16     };
17
18     rListConfiguration<Component> config = new rListConfiguration<> (
19         "Test-Komponenten",
20         new String[] { "Header1", "Header2" },
21         dummyRepo,
22         c -> new String[] { "dummy" }
23     );
24
25     InputReader reader = new InputReader();
26     ShowListCommand<Component> command = new ShowListCommand<>(config, reader);
27     command.render("Simulation");
28
29     System.setOut(originalOut);
30     String output = outContent.toString();
31     assertTrue(output.contains("Test-Komponenten"), "Die Komponententitel sollten ausgegeben werden.");
32 }
```

5. Unit Tests – Menu Tests

```
1  @Test
2      @DisplayName("Testet, ob ExitCommand das Runnable ausführt und 'Auf Wiedersehen!' ausgibt")
3      void exitCommandExecutes() {
4          final boolean[] wasRun = { false };
5          Runnable onExit = () -> wasRun[0] = true;
6
7          ByteArrayOutputStream outContent = new ByteArrayOutputStream();
8          PrintStream originalOut = System.out;
9          System.setOut(new PrintStream(outContent));
10
11          ExitCommand exitCommand = new ExitCommand(onExit);
12          exitCommand.execute();
13
14          System.setOut(originalOut);
15          String output = outContent.toString();
16
17          assertTrue(wasRun[0], "Das Runnable sollte ausgeführt worden sein.");
18          assertTrue(output.contains("Auf Wiedersehen!"), "Die Abschiedsnachricht sollte ausgegeben werden.");
19      }
```

5. Unit Tests – Menu Tests



```
1  @Test
2      @DisplayName("Testet, ob BackCommand ausführbar ist")
3      void backCommandExecutes() {
4          final boolean[] wasRun = { false };
5          Runnable onBack = () -> wasRun[0] = true;
6
7          BackCommand backCommand = new BackCommand(onBack);
8          backCommand.execute();
9
10         assertTrue(wasRun[0], "Das Runnable sollte ausgeführt worden sein.");
11
12     }
```


5. Unit Tests – Menu Tests



```
1  @Test
2      @DisplayName("Testet, ob ein Menü geöffnet werden kann")
3      void openMenuCommandOpensMenu() {
4          DummyMenu menu = new DummyMenu("TestMenu", new InputReader());
5          OpenMenuCommand command = new OpenMenuCommand(menu);
6          command.execute();
7          assertTrue(menu.opened, "Das Menü sollte geöffnet worden sein.");
8      }
```


5. Unit Tests – Database Tests



```
1  @Test
2      void userRepositoryCanSaveAndLoadUser() {
3          UserRepository userRepository = new UserRepository(connectionFactory);
4          String userName = "Integration Test User " + System.nanoTime();
5
6          User savedUser = userRepository.save(userName);
7          assertTrue(savedUser.getId() > 0);
8
9          User loadedUser = userRepository.findById(savedUser.getId());
10         assertNotNull(loadedUser);
11         assertEquals(savedUser.getId(), loadedUser.getId());
12         assertEquals(userName, loadedUser.getName());
13     }
```

5. Unit Tests – Database Tests

```
1  @Test
2  void computerRepositoryCanSaveAndLoadById() {
3      UserRepository userRepository = new UserRepository(connectionFactory);
4      ComputerRepository computerRepository = new ComputerRepository(connectionFactory);
5
6      User user = userRepository.save("Computer Repo Test User " + System.nanoTime());
7      Computer computer = createCompatibleComputer(connectionFactory);
8
9      int savedComputerId = computerRepository.save(user.getId(), computer);
10     assertTrue(savedComputerId > 0);
11
12     List<Computer> userComputers = computerRepository.findAllById(user.getId());
13     assertEquals(1, userComputers.size());
14
15     Computer loadedComputer = userComputers.get(0);
16     assertEquals(computer.getCPU().getId(), loadedComputer.getCPU().getId());
17     assertEquals(computer.getMainboard().getId(), loadedComputer.getMainboard().getId());
18     assertEquals(computer.getRAM().getId(), loadedComputer.getRAM().getId());
19     assertEquals(computer.getPSU().getId(), loadedComputer.getPSU().getId());
20     assertEquals(computer.getComputerCase().getId(), loadedComputer.getComputerCase().getId());
21 }
```

5. Unit Tests – Domain Tests



```
1  @Test
2      public void testGPUCreation() {
3          GPU gpu = new GPU(1, "Test GPU", 299.99, new Manufacturer(1, "Test Manufacturer"), 2220, 3290.0, 16, 190);
4          assertNotNull(gpu);
5          assertEquals("Test GPU", gpu.getName());
6          assertEquals(299.99, gpu.getPrice());
7          assertEquals("Test Manufacturer", gpu.getManufacturer().getName());
8          assertEquals(2220, gpu.getCoreClockMHz());
9          assertEquals(3290.0, gpu.getBoostClockMHz());
10         assertEquals(16, gpu.getVramGB());
11         assertEquals(190, gpu.getPowerConsumptionW());
12     }
```

5. Unit Tests – Computer Tests

```
1  @Test
2      public void testComputerCreation() {
3
4          PSUFormFactor psuFormFactor = new PSUFormFactor(0, "SFTX");
5          MotherboardFormFactor motherboardFormFactor = new MotherboardFormFactor(34, "ATX");
6          Socket socket = new Socket(1, "AM4");
7
8          Computer computer = new Computer.Builder()
9              .setCPU(new CPU(1, "Test CPU", 199.99, new Manufacturer(1, "Test Manufacturer"), socket, 8, false, 95))
10             .setGPU(new GPU(1, "Test GPU", 299.99, new Manufacturer(1, "Test Manufacturer"), 16, 150))
11             .setMainboard(new Mainboard(1, "Test Mainboard", 149.99, new Manufacturer(1, "Test Manufacturer"), socket, motherboardFormFactor, 4, 3, 6, 2))
12             .setRAM(new RAM(1, "Test RAM", 79.99, new Manufacturer(1, "Test Manufacturer"), 16, 15), 2)
13             .setPSU(new PSU(1, "Test PSU", 89.99, new Manufacturer(1, "Test Manufacturer"), 650, psuFormFactor))
14             .setComputerCase(new Case(1, "Test Case", 59.99, new Manufacturer(1, "Test Manufacturer"), motherboardFormFactor, psuFormFactor, true, 9))
15             .setStorageDevices(new Storage[] {
16                 new SSD(1, "Test SSD", 129.99, new Manufacturer(1, "Test Manufacturer"), 512),
17                 new HDD(2, "Test HDD", 79.99, new Manufacturer(1, "Test Manufacturer"), 1024)
18             })
19             .build();
20
21         assertNotNull(computer);
22         //computer.printConfiguration();
23     }
```

5. Unit Tests – Computer Tests



```
1  @Test
2      public void testWrongComputer() {
3          Computer computer = new Computer.Builder()
4              .setCPU(null)
5              .setGPU(null)
6              .setMainboard(null)
7              .setRAM(null, 0)
8              .setPSU(null)
9              .setComputerCase(null)
10             .setStorageDevices(new Storage[] {})
11             .build();
12
13         assertNull(computer);
14     }
```

5. Unit Tests – User Tests

```
1  @Test
2      public void loginTest() {
3          String username = "login-user";
4
5          String flowInput = String.join("\n",
6              "3", "1", username, "", "0", // Nutzer anlegen
7              "4", "1", username, "", "0", // Login -> Starten -> Username -> Enter -> Zurück
8              "0"                          // Beenden Hauptmenü
9          ) + "\n";
10
11         String output = runTestFlow(flowInput);
12
13         assertTrue(SessionManager.isLoggedIn());
14         assertEquals(username, SessionManager.getCurrentUser().getName());
15         assertTrue(output.contains("Willkommen zurück"));
16     }
17
```


5.1 ATRIP - Automatic

- Automatisches „Tear-Down“, sodass der Originalzustand wiederhergestellt wird



```
1  @AfterEach
2      public void tearDownEach() {
3          SessionManager.setCurrentUser(null);
4          System.setIn(originalIn);
5          System.setOut(originalOut);
6      }
```

5.1 ATRIP - Automatic

- Selbständige Bewertung der Tests durch Assertions

```
1  @Test
2  public void loginTest() {
3      String username = "login-user";
4
5      String flowInput = String.join("\n",
6          "3", "1", username, "", "0", // Nutzer anlegen
7          "4", "1", username, "", "0", // Login -> Starten -> Username -> Enter -> Zurück
8          "0"                          // Beenden Hauptmenü
9      ) + "\n";
10
11     String output = runTestFlow(flowInput);
12
13     assertTrue(SessionManager.isLoggedIn());
14     assertEquals(username, SessionManager.getCurrentUser().getName());
15     assertTrue(output.contains("Willkommen zurück"));
16 }
17
```

5.1 ATRIP – Thorough

- Testen von korrekten, als auch inkorrekten Eingaben

```
1  @Test
2  public void createCorrectComputer() {
3      String username = "builder-user";
4
5      String flowInput = String.join("\n",
6          "3", "1", username, "", "0", // Nutzer anlegen
7          "4", "1", username, "", "0", // Login
8          "2", "1", // Computerverwaltung -> Neu anlegen
9          "1", "1", "", "0", // CPU -> ID 1 -> Enter -> Zurück Paging
10         "2", "1", "", "0", // GPU -> ID 1 -> Enter -> Zurück Paging
11         "3", "1", "", "0", // Mainboard -> ID 1 -> Enter -> Zurück Paging
12         "4", "1", "4", "", "0", // RAM -> ID 1 -> 4 Module -> Enter -> Zurück Paging
13         "5", "1", "", "0", // PSU -> ID 1 -> Enter -> Zurück Paging
14         "6", "1", "", "0", // Case -> ID 1 -> Enter -> Zurück Paging
15         "7", "1", "", "0", // SSD -> ID 1 -> Enter -> Zurück Paging
16         "8", "", // Finish + waitForEnter
17         "0", // Zurück aus Konfigurator
18         "0", // Zurück aus Computerverwaltung
19         "0", // Beenden
20     ) + "\n";
21
22     String output = runTestFlow(flowInput);
23
24     User currentUser = SessionManager.getCurrentUser();
25     assertEquals(1, computerRepository.findAllByUserId(currentUser.getId()).size());
26     assertTrue(output.contains("ERFOLG"));
27 }
```

```
1  @Test
2  public void createWrongComputer() {
3      String username = "wrong-build-user";
4
5      String flowInput = String.join("\n",
6          "3", "1", username, "", "0", // Nutzer anlegen
7          "4", "1", username, "", "0", // Login
8          "2", "1", // Computerverwaltung -> Neu anlegen
9          "8", "", // Versuch speichern ohne Komponenten -> Print Error & Enter
10         "0", // Zurück Konfigurator
11         "0", // Zurück Computerverwaltung
12         "0", // Beenden
13     ) + "\n";
14
15     String output = runTestFlow(flowInput);
16
17     User currentUser = SessionManager.getCurrentUser();
18     assertTrue(computerRepository.findAllByUserId(currentUser.getId()).isEmpty());
19     assertTrue(output.contains("FEHLER"));
20 }
```

5.1 ATRIP – Thorough

pc-konfigurator

Element	Missed Instructions	Cov.	Missed Branches	Cov.	Missed	Cxty	Missed	Lines	Missed	Methods	Missed	Classes
de.ase.pcpartpicker.adapters.cli.commands		40 %		31 %	128	191	335	553	44	83	5	23
de.ase.pcpartpicker.adapters.cli		63 %		43 %	116	212	180	513	49	128	1	12
de.ase.pcpartpicker.part_assembly		74 %		55 %	93	165	86	324	4	40	1	6
de.ase.pcpartpicker.adapters.sqlite		84 %		60 %	86	181	103	506	3	50	0	4
de.ase.pcpartpicker.adapters.cli.utils		80 %		43 %	65	107	59	220	11	38	0	7
de.ase.pcpartpicker.adapters.sqlite.repositories		85 %		69 %	45	143	50	391	12	83	0	14
de.ase.pcpartpicker.domain		89 %		47 %	23	72	6	150	3	52	0	11
de.ase.pcpartpicker		0 %		0 %	9	9	17	17	8	8	2	2
de.ase.pcpartpicker.domain.HelperClasses		100 %		90 %	1	15	0	23	0	10	0	6
Total	3.919 of 13.716	71 %	590 of 1.194	50 %	566	1.095	836	2.697	134	492	9	85

5.1 ATRIP - Professional

- Test folgen denselben Prinzipien wie der restliche Code
 - Arrange, Act, Assert (Vorbereiten, Ausführen, Überprüfen)

```
1 @BeforeEach
2     public void setUpEach() {
3         originalIn = System.in;
4         originalOut = System.out;
5
6         userRepository = new InMemoryUserRepository();
7         computerRepository = new InMemoryComputerRepository();
8
9         SessionManager.setCurrentUser(null);
10        createCompatibleTestComponents();
11    }
```

```
1 @Test
2     public void loginTest() {
3         String username = "login-user";
4
5         String flowInput = String.join("\n",
6             "3", "1", username, "", "0", // Nutzer anlegen
7             "4", "1", username, "", "0", // Login -> Starten -> Username -> Enter -> Zurück
8             "0" // Beenden Hauptmenü
9         ) + "\n";
10
11        String output = runTestFlow(flowInput);
12
13        assertTrue(SessionManager.isLoggedIn());
14        assertEquals(username, SessionManager.getCurrentUser().getName());
15        assertTrue(output.contains("Willkommen zurück"));
16    }
17 }
```

```
1 @AfterEach
2     public void tearDownEach() {
3         SessionManager.setCurrentUser(null);
4         System.setIn(originalIn);
5         System.setOut(originalOut);
6     }
```

5.1 ATRIP - Professional

- Test folgen dem DRY Prinzip

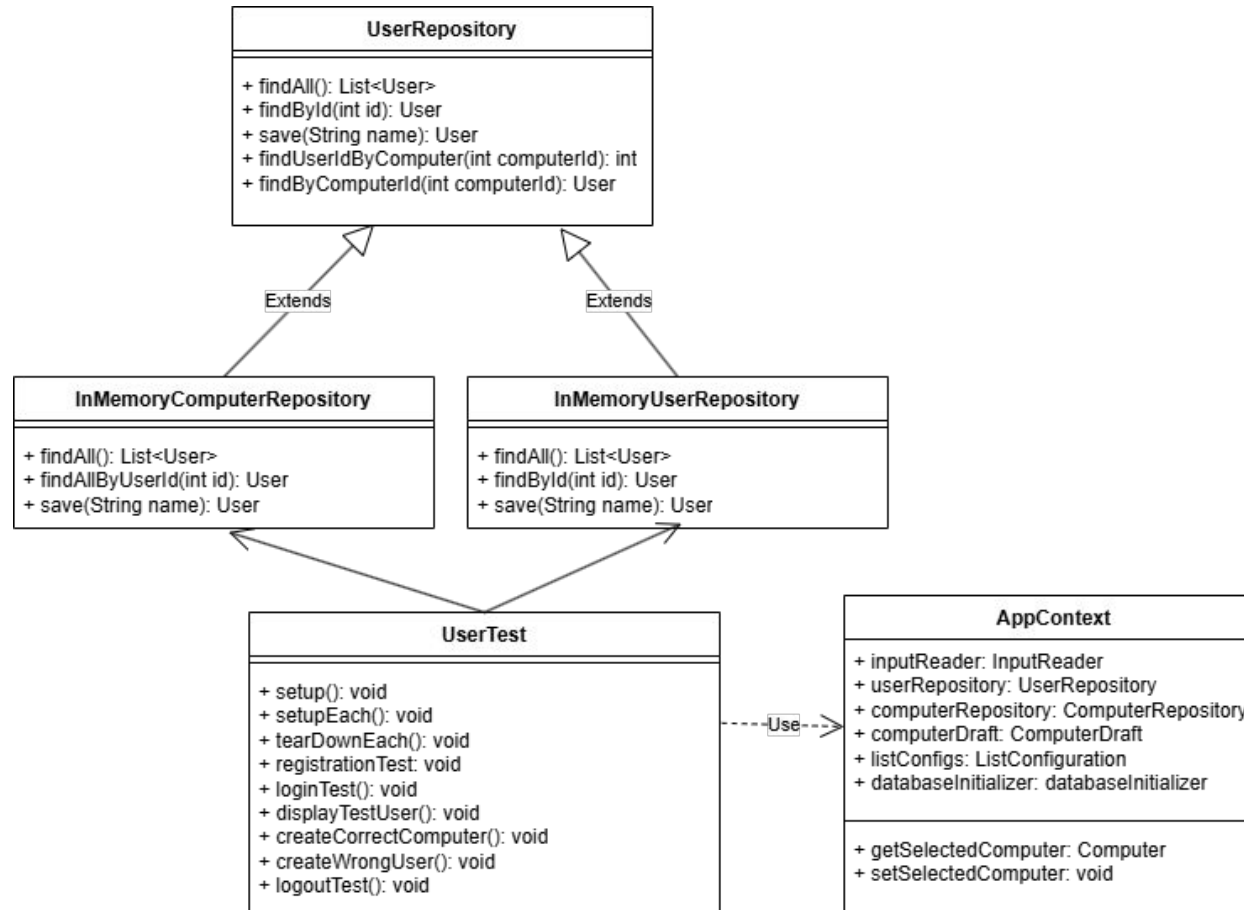
```
1 private String runTestFlow(String flowInput) {
2     ByteArrayOutputStream outContent = new ByteArrayOutputStream();
3     System.setOut(new PrintStream(outContent));
4     System.setIn(new ByteArrayInputStream(flowInput.getBytes(StandardCharsets.UTF_8)));
5
6     context = new AppContext();
7
8     try {
9         java.lang.reflect.Field userRepoField = AppContext.class.getDeclaredField("userRepository");
10        userRepoField.setAccessible(true);
11        userRepoField.set(context, userRepository);
12
13        java.lang.reflect.Field compRepoField = AppContext.class.getDeclaredField("computerRepository");
14        compRepoField.setAccessible(true);
15        compRepoField.set(context, computerRepository);
16
17        // WICHTIG: Sicherstellen, dass computerDraft im Context existiert
18        java.lang.reflect.Field draftField = AppContext.class.getDeclaredField("computerDraft");
19        draftField.setAccessible(true);
20        if (draftField.get(context) == null) {
21            draftField.set(context, new ComputerDraft());
22        }
23
24        // Sicherstellen, dass inputReader da ist
25        java.lang.reflect.Field readerField = AppContext.class.getDeclaredField("inputReader");
26        readerField.setAccessible(true);
27        if (readerField.get(context) == null) {
28            readerField.set(context, new InputReader());
29        }
30    } catch (Exception e) {
31        e.printStackTrace();
32    }
33
34    InputReader currentReader = null;
35    try {
36        java.lang.reflect.Field readerField = AppContext.class.getDeclaredField("inputReader");
37        readerField.setAccessible(true);
38        currentReader = (InputReader) readerField.get(context);
39    } catch (Exception e) {}
40
41    createMainMenu(currentReader).execute();
42
43    return outContent.toString(StandardCharsets.UTF_8);
44 }
```

5.1 ATRIP - Professional

- Lesbarkeit durch Kommentare erhöhen

```
1  @Test
2  public void loginTest() {
3      String username = "login-user";
4
5      String flowInput = String.join("\n",
6          "3", "1", username, "", "0", // Nutzer anlegen
7          "4", "1", username, "", "0", // Login -> Starten -> Username -> Enter -> Zurück
8          "0"                          // Beenden Hauptmenü
9      ) + "\n";
10
11     String output = runTestFlow(flowInput);
12
13     assertTrue(SessionManager.isLoggedIn());
14     assertEquals(username, SessionManager.getCurrentUser().getName());
15     assertTrue(output.contains("Willkommen zurück"));
16 }
17
```

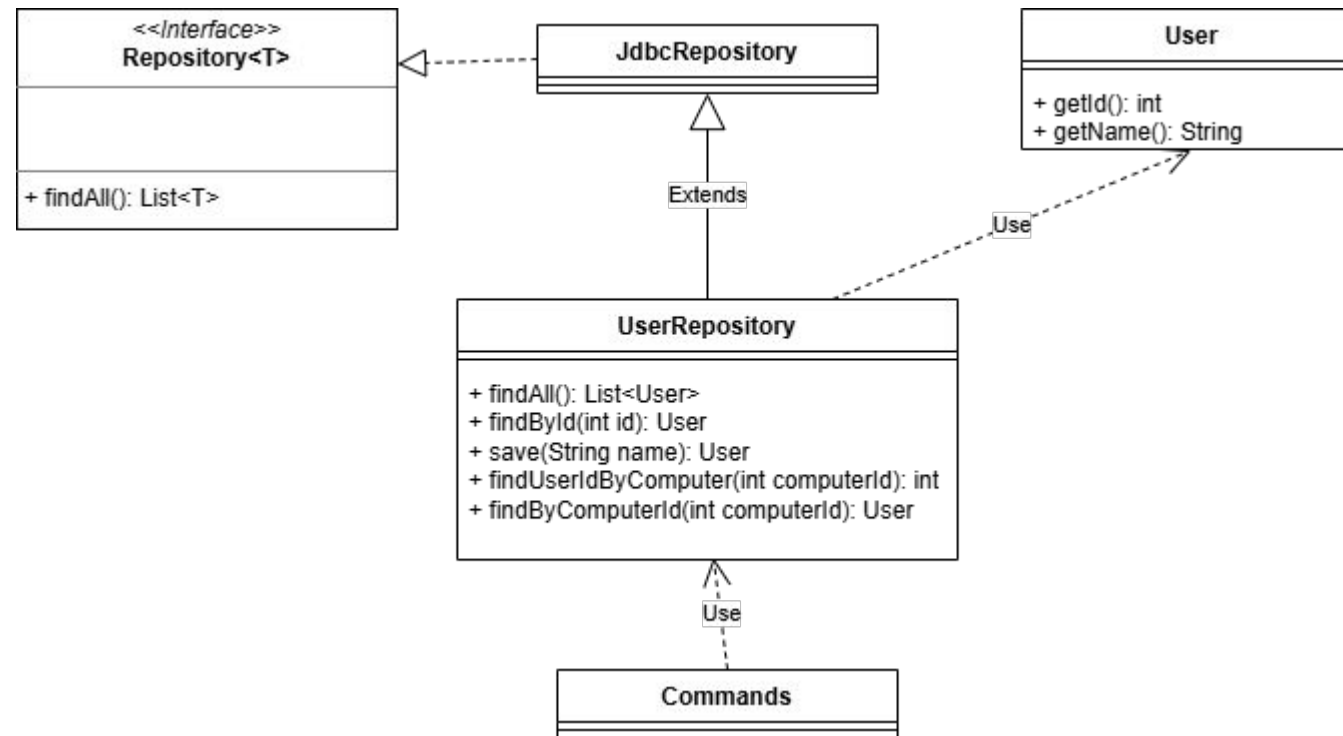

5.2 Fakes und Mocks - UML



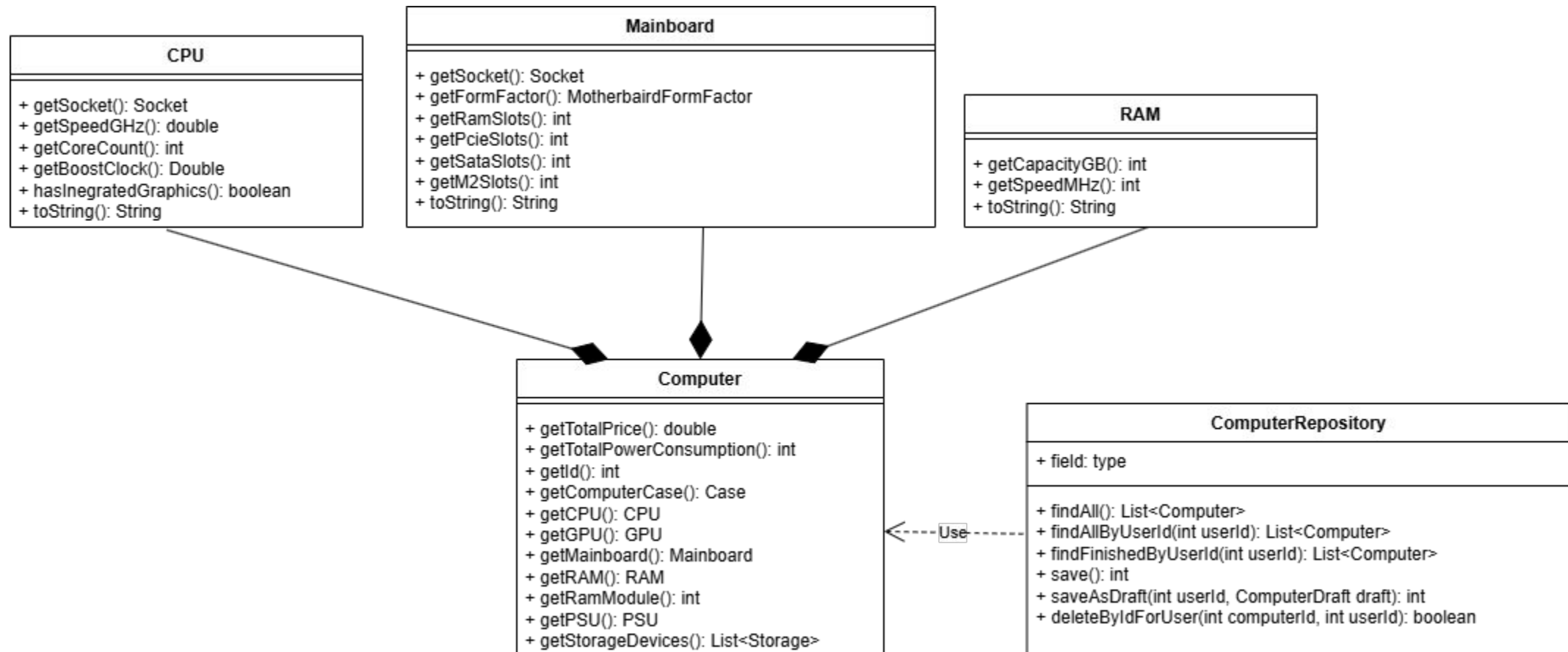
6. Domain Driven Design (DDD) - Ubiquitous Language

- ComputerDraft
- MotherboardFormFactor
- PowerConsumption
 - getWattage
- Bottleneck-Check

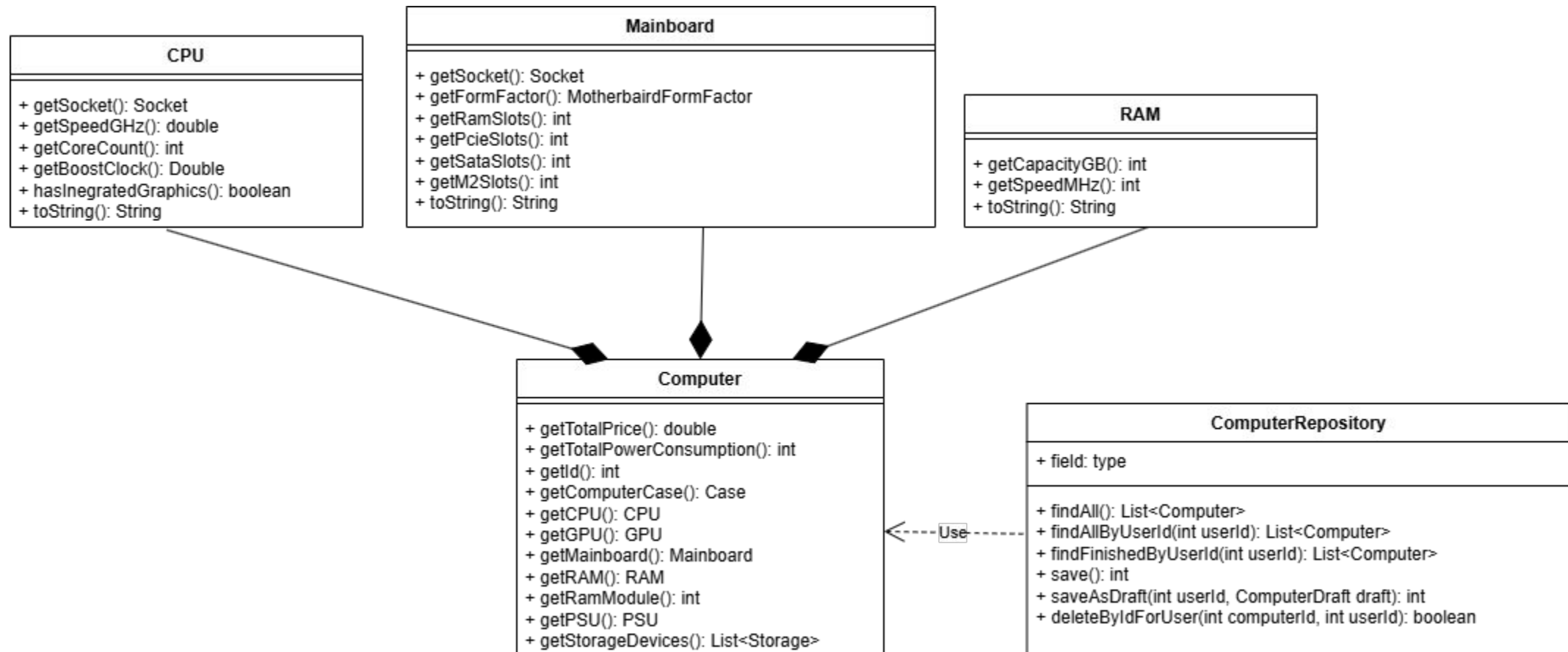
6.1 DDD – Repositories – UML



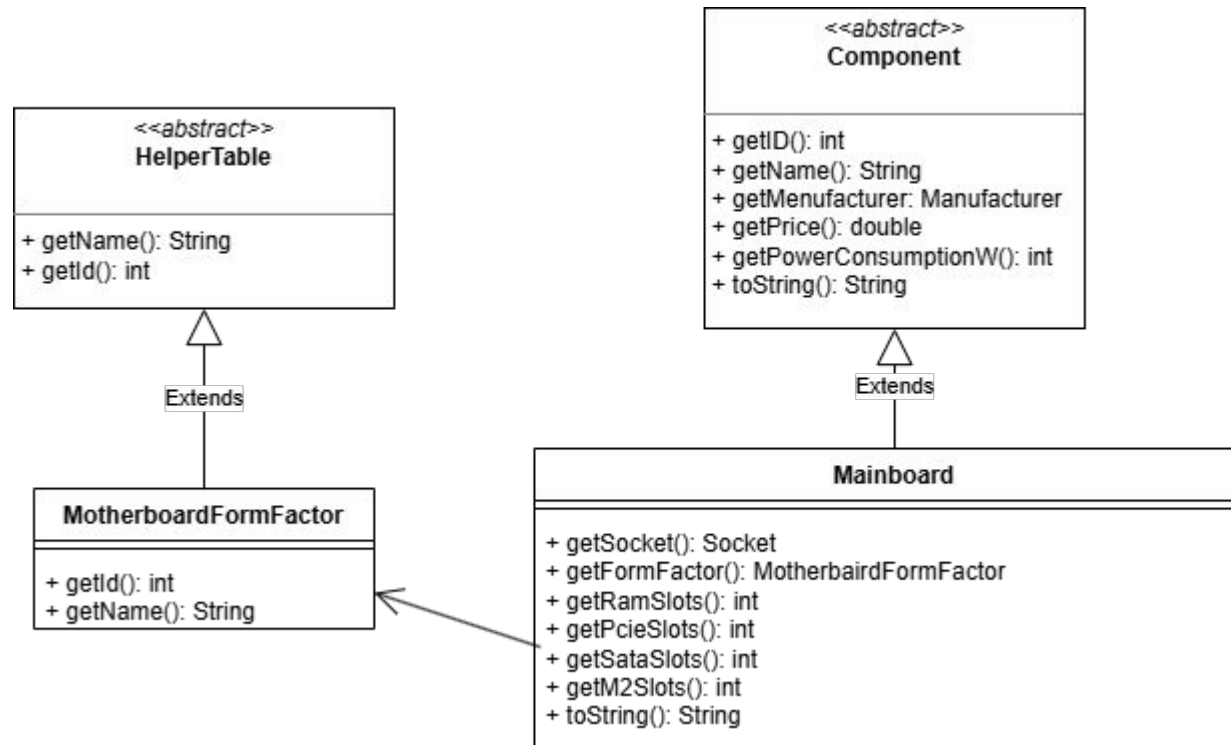
6.2 DDD – Aggregates



6.3 DDD – Entities



6.4 DDD – Value Objects



7. Refactoring – Code Smells – Long Method

```
1  /**
2   * Diese Methode überprüft die Kompatibilität der Komponenten.
3   * Es wird lediglich überprüft, ob der PC so gebaut werden und existieren könnte.
4   * Bspw. ob das Netzteil genug Leistung hat, um die Komponenten zu versorgen, wird hier nicht überprüft.
5   * @return true, wenn die Komponenten kompatibel sind und der Computer erstellt werden kann, andernfalls false. Es werden auch Fehlermeldungen ausgegeben, die erklären, warum der Computer nicht erstellt werden kann.
6   * @author Tizluhan
7   */
8  private boolean validate() {
9
10     if(ram == null) {
11         System.out.println(ColorConstants.RED("FEHLER") + " | Es muss mindestens ein RAM-Modul ausgewählt werden.");
12         return false;
13     }
14
15     if(cpu == null) {
16         System.out.println(ColorConstants.RED("FEHLER") + " | Es muss eine CPU ausgewählt werden.");
17         return false;
18     }
19
20     if(!cpu.hasIntegratedGraphics() && gpu == null) {
21         System.out.println(ColorConstants.YELLOW("WARNUNG") + " | Der Computer hat keine dedizierte Grafikkarte und die CPU verfügt nicht über integrierte Grafik. Es könnte zu Problemen bei der Anzeige kommen.");
22     }
23
24     if(mainboard == null) {
25         System.out.println(ColorConstants.RED("FEHLER") + " | Es muss ein Mainboard ausgewählt werden.");
26         return false;
27     }
28
29     if(psu == null) {
30         System.out.println(ColorConstants.RED("FEHLER") + " | Es muss ein Netzteil ausgewählt werden.");
31         return false;
32     }
33
34     if(computerCase == null) {
35         System.out.println(ColorConstants.RED("FEHLER") + " | Es muss ein Gehäuse ausgewählt werden.");
36         return false;
37     }
38
39     if(!computerCase.getPSUFormFactor().equals(psu.getFormFactor())) {
40         System.out.println(ColorConstants.RED("FEHLER") + " | Die Form des Netzteils passt nicht zum Gehäuse.");
41         return false;
42     }
43
44     if(!checkMainboardCaseCompatibility(mainboard, computerCase)) {
45         System.out.println(ColorConstants.RED("FEHLER") + " | Das Mainboard passt nicht ins Gehäuse.");
46         return false;
47     }
48
49     if(!cpu.getSocket().equals(mainboard.getSocket())) {
50         System.out.println(ColorConstants.RED("FEHLER") + " | Die CPU ist nicht mit dem Mainboard kompatibel.");
51         return false;
52     }
53
54     if(ramModule > mainboard.getRamSlots()) {
55         System.out.println(ColorConstants.RED("FEHLER") + " | Das Mainboard hat nicht genug RAM-Slots für die Anzahl der RAM-Module.");
56         return false;
57     }
58
59     if(storageDevices.isEmpty()) {
60         System.out.println(ColorConstants.RED("FEHLER") + " | Es muss mindestens ein Speichermedium ausgewählt werden.");
61         return false;
62     }
63
64     if(psu.getWattage() < calculateMomentumaryPowerConsumption()) {
65         System.out.println(ColorConstants.RED("FEHLER") + " | Das Netzteil hat nicht genug Leistung um das System zu versorgen. Das Netzteil muss mindestens " + calculateMomentumaryPowerConsumption() + "W liefern, um alle Komponenten zu versorgen.");
66         return false;
67     }
68
69     return true;
70 }
```

7. Refactoring – Code Smells – Long Method - Lösung

```
1 private boolean validate() {
2     if (!hasAllRequiredComponents()) return false;
3     if (!areFormFactorsCompatible()) return false;
4     if (!isHardwareCompatible()) return false;
5     if (!hasSufficientPower()) return false;
6
7     printOptionalWarnings(); // z.B. Hinweis auf fehlende GPU
8     return true;
9 }
10
11 // --- Die extrahierten Hilfsmethoden ---
12
13 private boolean hasAllRequiredComponents() {
14     if (ram == null || cpu == null || mainboard == null || psu == null || computerCase == null || storageDevices.isEmpty()) {
15         System.out.println("FEHLER | Es fehlen zwingend erforderliche Komponenten.");
16         return false;
17     }
18     return true;
19 }
20
21 private boolean areFormFactorsCompatible() {
22     if (!computerCase.getPSUFormFactor().equals(psu.getFormFactor())) {
23         System.out.println("FEHLER | Die Form des Netzteils passt nicht zum Gehäuse.");
24         return false;
25     }
26     if (!checkMainboardCaseCompatibility(mainboard, computerCase)) {
27         System.out.println("FEHLER | Das Mainboard passt nicht ins Gehäuse.");
28         return false;
29     }
30     return true;
31 }
32
33 private boolean isHardwareCompatible() {
34     if (!cpu.getSocket().equals(mainboard.getSocket())) {
35         System.out.println("FEHLER | Die CPU ist nicht mit dem Mainboard kompatibel.");
36         return false;
37     }
38     if (ramModule > mainboard.getRamSlots()) {
39         System.out.println("FEHLER | Zu viele RAM-Module für dieses Mainboard.");
40         return false;
41     }
42     return true;
43 }
44
45 private boolean hasSufficientPower() {
46     if (psu.getWattage() < calculateMomentaryPowerConsumption()) {
47         System.out.println("FEHLER | Das Netzteil hat nicht genug Leistung.");
48         return false;
49     }
50     return true;
51 }
```

7. Refactoring – Code Smells – Long Parameter List

TableUtils.java



```
1 private static String[][] buildTableRows(Case pcCase, CPU cpu, GPU gpu, Mainboard mb, RAM ram, int ramModule, PSU psu, List<Storage> storage, int totalPower, double totalPrice) {  
2
```

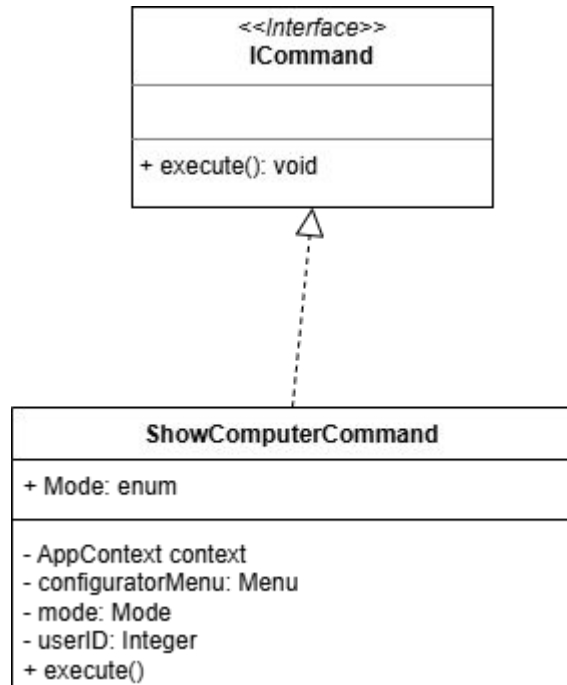

7. Refactoring – Code Smells – Long Parameter List – Lösung

```
1 public interface IComputerData {
2     Case getComputerCase();
3     CPU getCPU();
4     GPU getGPU();
5     Mainboard getMainboard();
6     RAM getRAM();
7     int getRamModule();
8     PSU getPSU();
9     List<Storage> getStorage();
10    int getTotalPowerConsumption();
11    double getTotalPrice();
12 }
13
14 public class Computer implements IComputerData { /* ... */ }
15 public class ComputerDraft implements IComputerData { /* ... */ }
16
17 public class TableUtils {
18
19     public static String[][] getComputerAsTableRows(Computer computer) {
20         return buildTableRows(computer);
21     }
22
23     public static String[][] getDraftAsTableRows(ComputerDraft draft) {
24         return buildTableRows(draft);
25     }
26
27     private static String[][] buildTableRows(IComputerData data) {
28         List<String[]> rows = new ArrayList<>();
29
30         if (data.getCPU() != null) {
31             rows.add(new String[]{"CPU", "Name", data.getCPU().getName()});
32         }
33
34         return rows.toArray(new String[0][]);
35     }
36 }
```

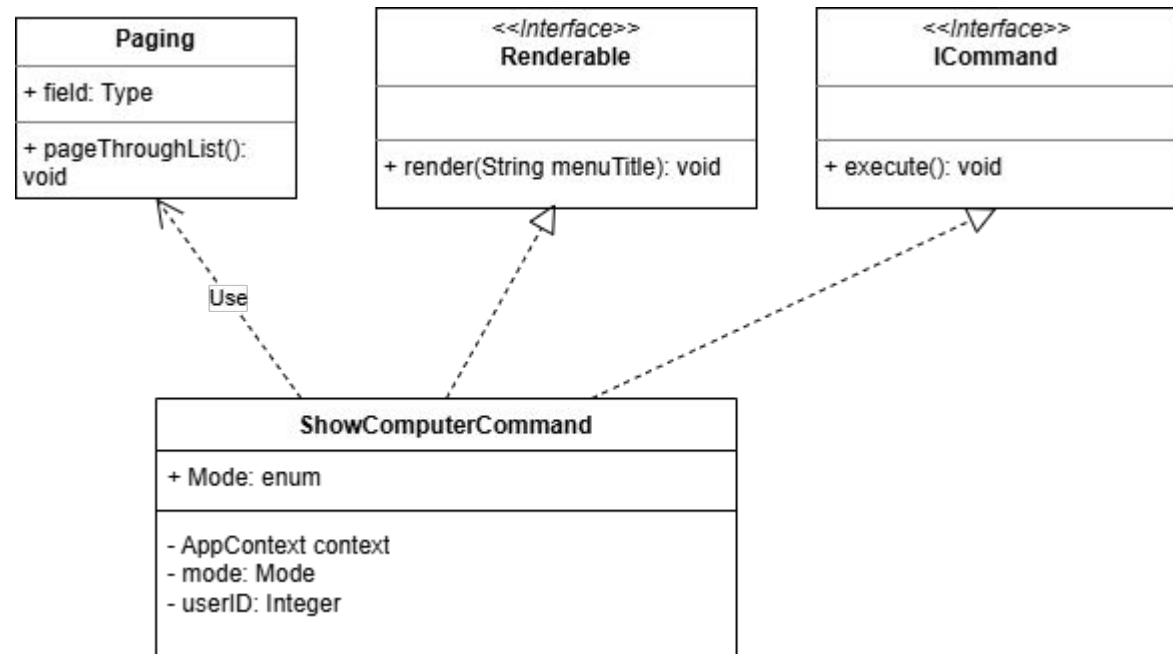
7.1 Refactoring – Extract Class – UML

Commit: fed4d10

Vorher

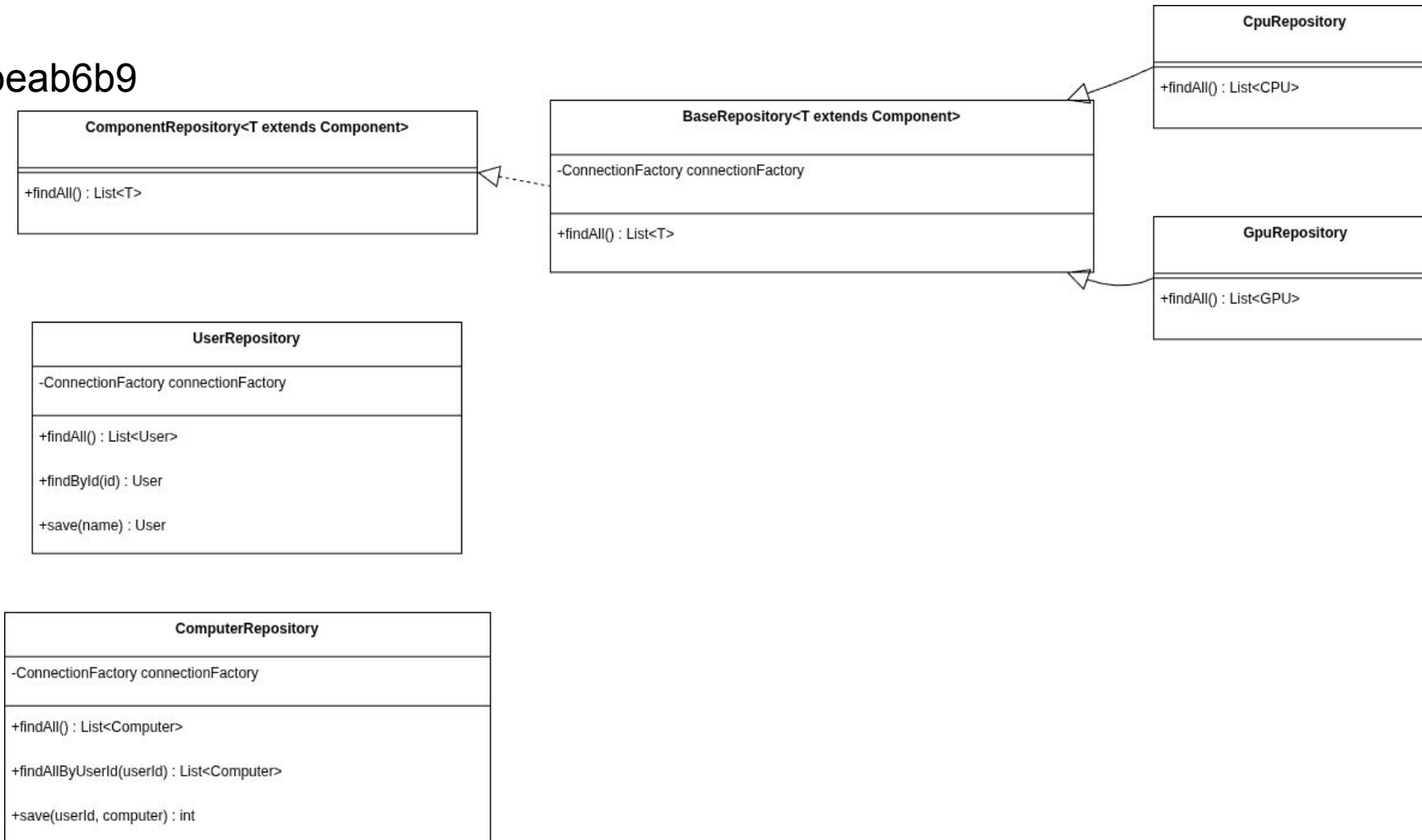


Nachher

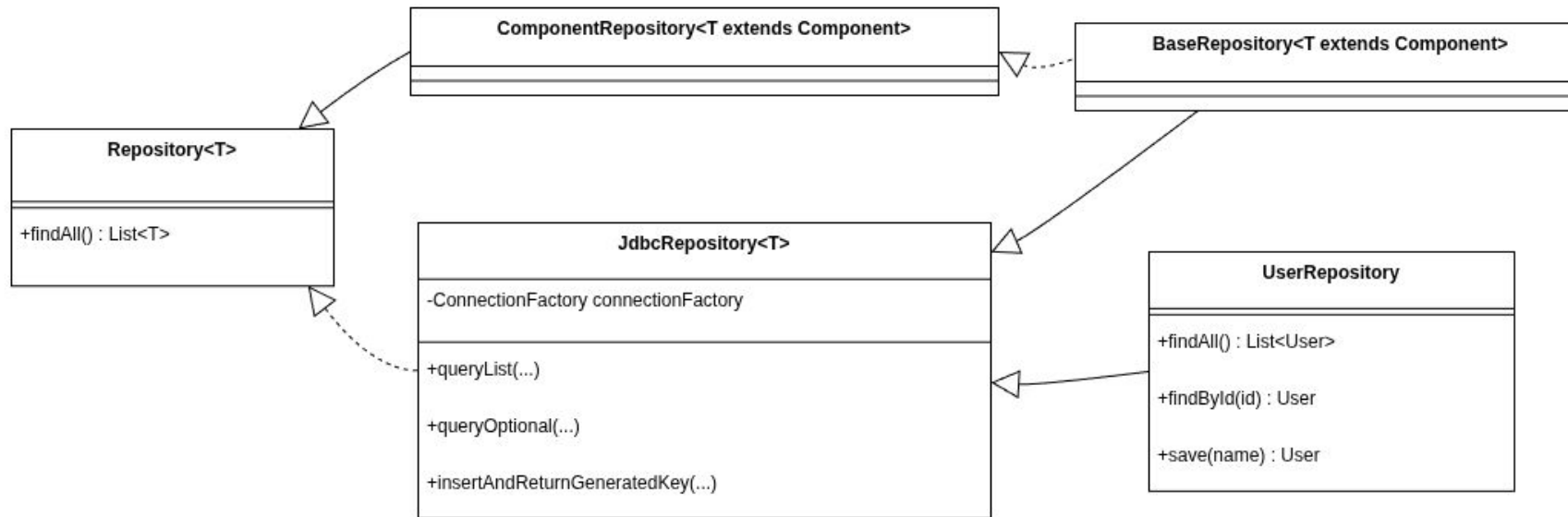


7.2 Refactoring - Duplicated Code (vorher)

Commit: beab6b9



7.2 Refactoring - Duplicated Code (nachher)

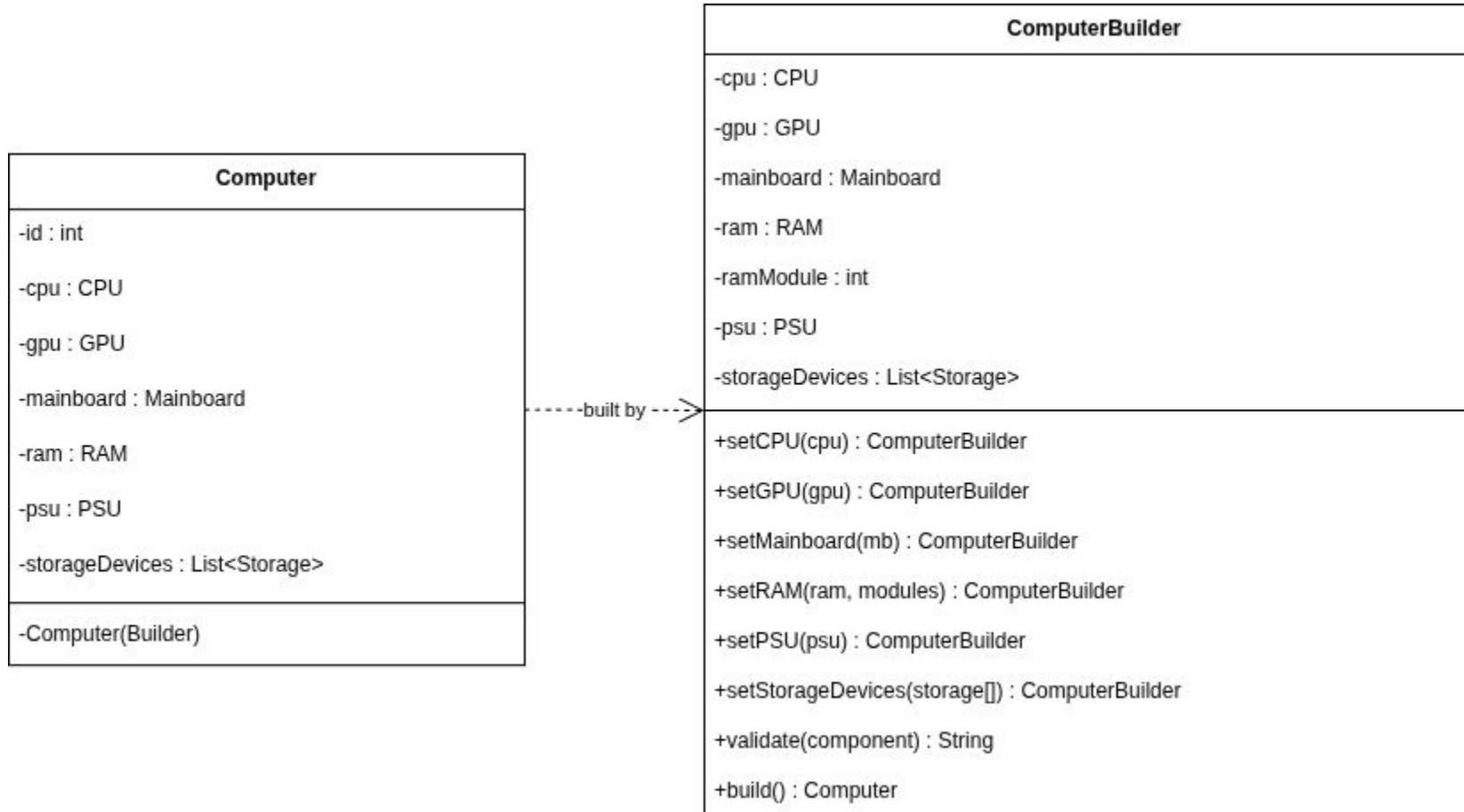


8. Entwurfsmuster - Builder

```
public static class Builder {  
    private CPU cpu;  
    private GPU gpu;  
    private Mainboard mainboard;  
    private RAM ram;  
    private int ramModule;  
    private PSU psu;  
    private List<Storage> storageDevices = new ArrayList<>();  
  
    public Builder setCPU(CPU cpu) { this.cpu = cpu; return this; }  
    public Builder setGPU(GPU gpu) { this.gpu = gpu; return this; }  
    public Builder setRAM(RAM ram, int modules) { this.ram = ram; this.ramModule = modules; return this; }  
  
    public String validate(Component component) { ... }  
  
    public Computer build() { ... }  
}
```

```
// Verwendung im Ablauf  
Computer newComputer = context.computerDraft.getBuilder().build();  
int saveId = context.computerRepository.save(currentUser.getId(), newComputer);
```

8. Entwurfsmuster - Builder - UML



8. Entwurfsmuster - Composite

```
public interface IMenuComponent {
    void execute();
    String getTitle();
    default boolean isVisible() { return true; }
}

public class MenuItem implements IMenuComponent {
    private final ICommand command;

    @Override
    public void execute() {
        command.execute();
    }
}

public class Menu implements IMenuComponent {
    private final List<IMenuComponent> children = new ArrayList<>();

    public void add(IMenuComponent component) {
        children.add(component);
    }

    @Override
    public void execute() {
        ...
        for (IMenuComponent child : children) {
            if (child.isVisible()) {
                System.out.println(child.getTitle());
            }
        }
        ...
    }
}
```

```
// Aufbau in MenuFactory
Menu mainMenu = new Menu("PC Part Picker - Hauptmenü", context.inputReader);
mainMenu.add(new MenuItem("Computerverwaltung", new OpenMenuCommand(createComputerMenu())));
mainMenu.add(new MenuItem("Nutzerverwaltung", new OpenMenuCommand(createUserMenu())));
```

8. Entwurfsmuster - Composite - UML

